Entry and Pricing with Fighting Brands: Evidence from Generics and Authorized Generics*

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Abstract

We study generic entry decisions when facing both uncertain approval times and entry-deterring strategies by the incumbent, specifically the release of an Authorized Generic (AG). Using sales and revenue data on the US, we estimate a structural model of generic entry and timing of AG release, then simulate counterfactuals. Generics and AGs deter each other differently - generics can apply for entry earlier, but AGs can launch earlier. Faster generic approval leads to weakly fewer generic entrants, lower prices in the early periods of the market, and weakly higher prices in later periods. A ban on AG leads to weakly more generics, and often multiple new generics. The effect on prices is mixed - while there is increased price competition from more generics in the later stages of the market, it is counteracted by lack of an additional competitor through the AG in the early stages of the market.

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1 Introduction

We study firms' entry decisions when facing both uncertain regulatory approval times and entry-deterring strategies from the incumbent. Specifically, we look at generic drug manufacturers' entry decisions into molecule-formulations. Generics need to get approval from FDA to launch; however, the time to get approval is uncertain and lengthy.[1] In addition, they face strategic responses from the branded drug manufacturer that deter or blunt generic competition. One such strategy is for the brand manufacturer to release an Authorized Generic (AG). AGs are chemically identical to the branded drug but without the brand label attached. These are priced substantially lower than the brand and close to generics' prices. This (potential) additional competition from a similar product worsens the prospects of generics considering entry into the market. In the marketing literature this is known as a "fighting brand strategy".

In this paper, we shed light on three issues. First, we study the economics of entry decisions when facing both uncertain approval times and entry-deterrence through fighting-brand launches. Second, we examine the effect of faster approval times on equilibrium market outcomes. Third, we conceptualize the impact of AGs on short-term and long-term market outcomes, particularly generic entry deterrence.

To do so, we develop a stylized structural model of i) generics making static entry decisions, and ii) the branded incumbent making a dynamic timing decision to release an AG. This model guides us through the key economic concepts at work and how these interact. We perform counterfactual simulations to show the following. First we study how generics and AGs can deter each other. Because generics move first, they can apply for entry earlier; however, since AGs do not need regulatory approval, they can deter generics by being able to launch their products into the market earlier. Second, a faster approval rate leads to weakly *fewer* generic entrants, lower prices in the early periods of the market, and weakly higher prices in later periods of the market. Finally, we show that a ban on AG leads to weakly more generic entrants. In particular, banning AGs can lead to *multiple* additional generics entering the market. The AG ban generally leads to higher prices in the early periods of the market. Prices are lower in the later periods of the market if the AG

[1] In the past, slow approval rates had been blamed for persistent high drug prices after loss of market exclusivity as generics wait to launch. The FDA has taken steps to speed up the approval process. See Section — for a detailed discussion.

would have deterred multiple generics. The takeaway from simulating the impact of an AG ban is that while there is increased price competition from weakly more generic entrants in the later stages of the market, it is counteracted by lack of additional price competition from the AG in the early stages of the market.

The setting and our contributions are useful for several reasons. First, since generic competition is a key driver in lowering drug prices and spending, understanding the economic mechanisms at play is important for crafting policies. Second, while our structural model is strictly about AGs, we believe it can be extended to study other strategic reactions against generics by branded drug manufacturers (product line extensions, REMS requirements, orphan drug classification, etc.) and potentially additional regulatory frictions. The model can also be used to study fighting brand strategies outside pharmaceuticals. Finally, fighting brands are widely observed in many industries besides pharmaceuticals, so this study adds to our understanding of such marketing strategies, particularly entry deterrence motives.[2]

An outline of our stylized structural model is as follows. The demand side is a discrete choice model where consumers first choose a group - brand, nonbrand (generics and AG), or outside option - then choose a product within the group. The supply side of the industry is modeled as two separate stages. In the first stage, generic manufacturers make a static entry decision on whether to enter a molecule-formulation market. In the second stage, a dynamic game begins where every period, generics who decided to enter are randomly approved for product launch by the FDA, the branded drug manufacturer decides whether to release an AG, and market participants compete on prices. This two-stage model setup is because unlike generics, AGs do not need FDA approval to launch. We solve the two-stage supply model by backward induction, and as a result allow for AG and generics to form expectations about each others' entry and pricing decisions when making a choice.

We use data on national sales and revenue for prescription drugs in the US between

[2]When facing new entrants, incumbents sometimes engage in price discrimination through release of fighting brands. In a fighting brand strategy, the incumbent releases a low brand-value version of its existing product, called a fighting brand. The high brand-value product charges high price and the fighting brand charges low price. This segments the market, with fighting brand competing with new entrants and original product serving the higher end of the market. Examples of fighting brands (from Bourreau et al. (2021)) include: Intel with the Pentium series (brand) and Celeron (fighter brand) to compete with AMD; Lufthansa (brand) with a low-cost subsidiary Germanwings (fighter brand) to fight against low-cost carriers; Canadian telecom providers Rogers (brand) marketing low-cost alternative (Chatr).

2004-2016 (IQVIA National Sales Perspective). The demand model is estimated using the methods of Berry et al. (1995) and Maggio et al. (2022). Our demand estimation recovers the price sensitivity of consumers and their preferences for branded and non-branded drugs. Our supply side estimation pins down the entry cost distribution of AGs and variance of choice-specific shocks. Prices in different industry states are predicted using a reduced-form regression to avoid making pricing conduct assumptions. Following previously published papers, our market definition is the molecule-formulation pair (Conti and Berndt (2020), Wang et al. (2022), Yurukoglu et al. (2017)).

For our counterfactuals, we solve the model for different observed market specifications. An important element of model dynamics is that the early quarters of a market are most profitable for generics and AGs – there are few generics present as the rest wait to get FDA approval, resulting in lower price competition and high markups.

Our first set of counterfactuals illustrate how both generics and AGs can deter each other's entry, but in different ways. Generics can deter AG because they move first, i.e. they apply for entry and incur the entry cost first. However, AGs can deter generics because they can launch their product earlier than generics – the early launch significantly reduces generics' expected profits by providing price competition in the early profitable periods of the market. To showcase these two forces, we first show that no AG is released when the generics can launch immediately and the AG has no cost or demand advantage; we then show that AG is able to deter more generics when it can launch earlier rather than being forced to delay.

Our second set of counterfactuals study how faster generic approval by the FDA affects market outcomes. Faster generic approval has been a priority of the FDA since the 2010s (Berndt et al. (2018)). We find that a faster approval rate leads to weakly fewer generic entrants, lower prices in the early periods of the market, weakly higher prices in the later periods of the market, and that the effect is more muted if AGs may be present in the market. Why does a faster generic approval rate lead to lower generic entry? While a faster approval rate means a longer stream of revenue accruing to a generic from earlier launch, it is counteracted by greater presence of rivals upon launch. The latter means greater price competition and lower profits upon launch. Greater competition upon launch offsets the advantage from an early launch, reducing lifetime generic profits and therefore weakly reducing generic entry.

Our third set of counterfactuals study how the presence of AGs affects short-term and long-term market outcomes compared to an absence of AGs. We show that a ban on AG leads to usually leads to more generic entrants, and in fact may lead to multiple additional generics entering the market (40% of market specifications). The AG ban can incentivize multiple new generic entrants because it is a more potent competitor than independent generics as a result of its early-launch advantage, and so has a more negative impact on generics' expected profits. If the AG is banned, not only is there one fewer competitor, but it is also one less competitor in the most profitable periods in the market.

The effect on prices from the AG ban is mixed. When an AG is released, it is usually in the early periods of the market just after loss-of-exclusivity. This means that generics entering the market early will face an additional competitor through the AG. Therefore, the AG keeps prices low in the early periods of the market.[3] However, if the AG deters multiple generics from entering, then prices in the later periods of the market would be higher with the AG due to missing generic competition.

Contributions and related literature: This paper contributes to the following strands of literature. First, we contribute to a small literature on Authorized Generics. We are the first to study the impact of AGs on generic firms by using a structural model of entry. Moreover, our model incorporates this interaction in a rational expectations framework. This allows us to trace out feedback between AG and generic decisions, and in particular allows us to study how the presence/absence of AG affects generics' decisions. Furthermore, our model allows us to explore a wider variety of economic effects from the presence of the AG. A few papers have used reduced-form evidence to study Authorized Generics. Notably, Appelt (2015) uses a recursive bivariate probit regression to show that AG entry does not impact generic entry in Germany. An important source for us is a report from the Federal Trade Commission (FTC (2011)) that leveraged many information sources to explain the decision to release the AG and generic manufacturers' reactions to it. FTC (2011) mostly focuses on how AG release may disincentivize Paragraph IV lawsuits by generics and does so by compiling industry data and company documents; this paper instead focuses on how AGs and generics interact when a Paragraph IV lawsuit has not been filed.

[3] One exception is markets where the AG would have chosen to enter later. In such markets, the AG ban has minimal effect on early-period prices since AG is absent during that period, and in fact may even lower prices if an additional generic enters and receives early FDA approval.

Both FTC (2011) and this paper finds that the majority of AGs are released in the absence of a Paragraph IV lawsuit. A recent paper by Fowler et al. (2023) uses industry data to describe broad trends about Authorized Generics over 2010-2019, most of which matches our findings; see Appendix F for a comparison.

Second, we add to the literature on generic drug entry, particularly generic entry deterrence. While many papers have studied generic entry in the US, few use structural methods to model such decisions. Doing so allows us to see how changing key economic parameters affects entry incentives by generics. Furthermore, we contribute to the literature developing a structural model of generic entry deterrence that may be used to study other generic-deterrence strategies. An important paper for our purposes is Ching (2010), who studies generic entry and brand's dynamic pricing in 1984-1990 to model learning dynamics. We adopt part of our entry model from this paper. Other notable papers are Gallant et al. (2017), Morton (1999), Reiffen and Ward (2005), and Starc and Wollmann (2023). A paper that we think of as closely complementing ours is Wang et al. (2022), which uses a static entry model and reduced-form profit functions to understand the factors that promote generic entry into a molecule-formulation.[4]

Third, we contribute to a very sparse Empirical IO literature on fighting brands. We are one of the very few papers to build and estimate a model of an incumbent releasing a fighting brand. There is a theoretical literature on fighting brands, notably Anderson and Dana (2009), Deneckere and McAfee (1996), Johnson and Myatt (2003), and Nocke and Schutz (2018). An important empirical paper studying fighting brands is Bourreau et al. (2021). They show that in the French mobile telecommunications market, releasing fighting

[4]Our papers each focus on one aspect of the industry while abstracting away from another. In contrast to our paper, Wang et al. (2022) capture heterogeneity on the generics side, look at more molecule-formulations, and consider markets where generics never entered. In particular, they investigate how firm-specific covariates affect entry decisions, while our paper instead assumes that generics are homogeneous. On the other hand, unlike our paper, they i) do not explicitly account for dynamics, ii) do not allow for authorized generics, and iii) do not estimate a demand system and price competition (instead using a reduced-form expression to proxy for per-period profits). Our paper can track how market shares and prices change as the competitive landscape or policy environment changes, and use it to illustrate the knock-on effect on entry behavior. Furthermore, by accounting for dynamics, we can better simulate the impact of dynamic changes like faster FDA approvals and bans. As a result, we believe we are better positioned to answer how the number of generic entrants change due to faster FDA approval rates, something Wang et al. (2022) study, and we reach the opposite conclusion to their paper. In short, Wang et al. (2022) is better for understanding how firm-specific heterogeneity affects generic entry decisions into a wide range of markets, while our paper is better for thinking about market dynamics and authorized generics.

brands occurred due to a breakdown of collusion, and use a structural model of demand and supply to study this phenomenon.

Fourth, we construct a parsimonious structural model of pharmaceutical product entry that is quick to solve and can be easily extended in various directions (such as consumer heterogeneity, firm heterogeneity, presence of intermediaries, etc.) in future research. In doing so, we contribute to an extensive literature on static entry games and dynamic single-agent and oligopoly games. For static entry, key references include Berry (1992), Berry et al. (2016), Bresnahan and Reiss (1991), Mazzeo (2002), and Seim (2006). For dynamic single-agent and oligopoly games, some important references are Aguirregabiria and Mira (2007), Benkard (2004), Collard-Wexler (2013), Gowrisankaran and Rysman (2012), Igami (2017), Igami (2018), Igami and Uetake (2020), Ericson and Pakes (1995), Pakes et al. (2007), Rust (1987), and Ryan (2012); see Aguirregabiria et al. (2021) for a detailed overview of the literature. We adopt part of our model and estimation method from Igami (2018).

Fifth, we contribute to a somewhat limited empirical literature on entry deterrence and preemption. Unlike most papers in this area, we tackle empirical issues regarding entry deterrence differently. The fundamental empirical problem is that when a potential entrant does not enter, the researcher cannot tell if this was due to deterrence or due to absence of entry threat to begin with (see Cookson (2017) and Gil et al. (2024) for a detailed discussion and creative workarounds). In our paper, we tackle this issue by building a model of entrant's choices, and using it to predict entry decisions in different markets. The sequential nature of moves in this setting (i.e. generics make entry decision first followed by AG) leads to a clean solution of the entry deterrence problem. Empirical papers on this topic include Cookson (2017), Fang and Yang (2024a), Fang and Yang (2024b), Gil et al. (2024), Goolsbee and Syverson (2008), Hünermund et al. (2014), Igami and Yang (2016), Schmidt-Dengler (2024), and Seamans (2012). There is an extensive theoretical literature on this topic, which can be found in the references above.

Finally, our paper relates to a large literature on pharmaceuticals. A vast amount of work has been done on the theoretical and empirical side of this industry. Frank and Salkever (1992) was the first to lay out a theoretical model for why branded drug prices often stayed above generic prices. A non-exhaustive list of references that informed the current paper include Arcidiacono et al. (2013), Berndt et al. (2007), Bhattacharya and Vogt

(2003), Bokhari and Fournier (2013), Bokhari et al. (2020), Dubois et al. (2022), Dubois and Majewska (2022), Ellison and Ellison (2011), Frank and Salkever (1997), Ganapati and McKibbin (2023), Grennan et al. (2021), Hermosilla and Ching (2023), Lakdawalla (2018), Olson and Wendling (2018), Olssen and Demirer (2199), Reiffen and Ward (2007), Tenn and Wendling (2014), and Tunçel (2024); see Morton and Kyle (2011) for an overview. More specifically, we contribute to the literature of using structural models to understand the economics of the pharmaceutical industry; see Ching et al. (2019) for an overview. Berndt et al. (2017), Berndt et al. (2018), and Conti and Berndt (2020) provide important descriptive evidence about the aggregate dynamics and inner workings of the industry which we use to motivate our model.

2 Institutional Background

After the branded drug manufacturer loses market exclusivity, generics enter the market by applying for FDA approval. However, gaining FDA approval is lengthy, costly, and takes an uncertain amount of time. The brand drug manufacturer can respond by releasing an authorized generic (AG). The AG is chemically identical to the brand drug but without the brand label attached. Unlike generics, the AG can be launched at any time without having to wait for FDA approval. The general patterns after loss of exclusivity are: branded drug prices stay elevated, generics and AG charges low price which further declines with more competition, and the branded drug loses most of its market share to generics and AGs over time.

2.1 Generic Drugs and FDA approval

A pharmaceutical manufacturer invests in R&D to discover a drug, then patents the molecule and conducts tests for efficacy and safety. This drug is commonly referred to as the "branded drug" (see Dubois et al. (2015) and Khmelnitskaya (2023), and the references therein). Throughout we refer to the manufacturer that pioneered the molecule as the "branded drug manufacturer" and the pioneer molecule as the "branded drug". We use the term "nonbrand" to indicate both generics and AGs.

After loss of exclusivity, generic drugs enter the market by applying for approval from the FDA. The approval process is costly for the generic manufacturer, lengthy, and has uncertain duration.

To gain FDA approval, generic manufacturers need to prove to the FDA that their products are "bioequivalent" to the branded drug and can be produced at scale. Bioequivalence is defined to mean "product has the same active ingredient, dosage form, strength, route of administration and conditions of use" as the reference branded drug.[5] Generic manufacturers have to give further evidence that they can manufacture the drug accurately, at scale for commercial distribution, and according to Good Manufacturing Practices.[6] This might further involve a back and forth between the FDA and generic applicant until the FDA is satisfied with the production line.[7] An important aspect of proving bioequivalence is that the generic firm needs to reverse-engineer the branded drug. Since some aspects of the manufacturing process for the branded drug may be undocumented or kept as trade secrets by the branded manufacturer, generic firms must experiment with production to meet regulatory requirements. Estimates of the cost of ANDA application range from \$2 million to \$20 million.

The generic manufacturer then needs to submit an Abbreviated New Drug Application (ANDA) with all the relevant information attached. The mean approval time for an ANDA is between 32-40 months for our data period. Approval times of generics vary due to the amount of work involved in verifying the evidence submitted in support of the entrant's ANDA, as well as the FDA's workload. In the past several decades, the FDA has experienced a significant backlog of pending ANDA applications. Policies such as the GDUFA – and its reauthorization every 5 years – have given the FDA more resources to reduce the time and increase the scope of generic drug approvals (Berndt et al. (2018), Wang et al. (2022)).

In this paper, we refer to applying for ANDA approval as "generic entry", and refer to generics marketing their product after FDA approval as "product launch". Thus, generics

[5]Source: https://www.fda.gov/media/71401/download

[6]See information from the FDA on Good Manufacturing Practices: www.fda.gov/drugs/pharmaceutical-quality-resources/current-good-manufacturing-practice-cgmp-regulations.

[7]Bioequivalence ensures that generic drugs are not lower quality compared to their equivalent branded drugs - by definition, their bioequivalence means that they can be substituted with the branded product and have the same effect. Thus, generic drugs compete with their branded equivalents in the US on the basis of their availability and price.

enter a market by applying for FDA approval, and *launch* their product after receiving FDA approval.

2.2 Authorized Generics

In response to generic entry, the branded drug manufacturer can release an Authorized Generic (AG). The AG is chemically identical to the branded drug in terms of molecule structure and formulation, except that it doesn't have the brand label attached.

A key detail about product launch in this setting is that, unlike generics, AGs can be introduced anytime and without FDA approval. This is because they are riding on the branded drug's approval from the FDA.

Why would a branded drug manufacturer choose to not release an AG? There are several reasons. First, the AG could cannibalize the branded drug's sales. Second, the AG would lead to generics lowering their prices in response, further increasing the price gap between brand and generics products; this causes branded drug sales to fall. Finally, it could be due to supply side reasons such as entry costs. Fourth, it could be that many generics got FDA approval in the very first period after loss-of-exclusivity, meaning that the AG will make little profit. In our model we account for all these factors.

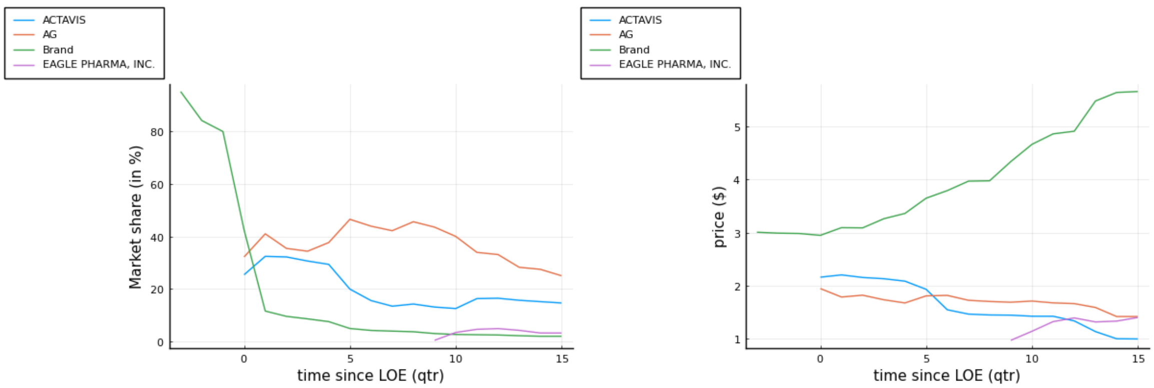
2.3 Evolution of Market Shares and Prices

In terms of market prices and shares, the general pattern we see in the data is as follows. After loss of market exclusivity by the branded drug, generics enter the market and undercut the brand's price. The branded drug's price typically remains stable or increases. An AG is sometimes released by a branded drug manufacturer in response to generic entry. The AG and generics price their products close to each other. Over time, the branded drug's market share dramatically falls, while those of nonbrand drugs rises. Examples of this pattern can be seen in Figure 1 for a molecule-formulation.

3 Data and Descriptive Statistics

We leverage data on quarterly sales and revenue of prescription drugs in the US from 2004-2016, and use this to construct market shares and prices. We define a market to be a

Figure 1: Evolution of market shares (left) and prices (right) for Diclofenac-Misoprostol (oral formulation).



molecule-formulation; and of the 241 molecule-formulations we restrict our attention to, about 45% see an AG released. Most AGs are released soon after the first generic entry occurs. Most markets have between 1-11 competitors.

3.1 Data

The data for this paper comes from IQVIA's National Sales Perspective™(NSP) database and covers sales in US for 2004 - 2016. NSP™ data contains national sales of prescription drugs to pharmacies, clinics, hospitals, and other distribution channels, aggregated to the quarterly level. Each observation includes the name of the molecule (active ingredient) and branded name (if relevant), quarter, sales amount in drug unit volume, sales amount in U.S. dollars, supplier of the product, distribution channel, product form (e.g., oral, injectable), and other information including therapeutic class, product launch date, etc. Following previous literature (Conti and Berndt (2020), Wang et al. (2022), Yurukoglu et al. (2017)), we refer to a drug as a group of products having the same molecule (or active ingredient) and product form.

For our purposes, a drug's *therapeutic class* describes what broad diagnoses it targets; a drug's *molecule* structure describes the active ingredients present; and a drug's *formulation* is its method of delivery, e.g. oral, injectable, etc.

We now give a brief overview of the NSP™ database; this closely follows Berndt et al.

(2017), which contains a more detailed description of the entire dataset.

The database is constructed through a projected audit that tracks drug sales from manufacturers or wholesalers to pharmacies and other distribution outlets. The measure of "revenue" for each drug is the total amount paid for the purchase of a product to the manufacturer.[8] Furthermore, the NSP database reports volumes of each drug sold in "standard units", where a standard unit is defined as the smallest unit of the drug standardized across dosage and therapeutic form (Dubois and Lasio (2018)). We compute prices as the ratio of total revenue to total quantity in standard units, following Dubois et al. (2022) among others. In addition, data on Authorized Generics and Paragraph IV lawsuits were hand-collected.[9] AGs were identified from FDA's maintained list of AGs, an online source maintained by some producers of AGs, and verifying with news reports on AG release. The list of molecules facing Paragraph IV lawsuits were collected from a publicly available list maintained by the FDA.[10]

To summarize, for each drug product we see quarterly sales in the US, revenue generated, list of active ingredients, formulation of the drug (oral, injectable, or other), and therapeutic class (ATC3). Sales for a drug-molecule-formulation are aggregated by dosage and strength. Finally, we define a market at the molecule-formulation level.[11]

[8]Notably, this measure of revenue does not include rebates that drug manufacturers give back to other intermediaries. Generic manufacturers do not pay rebates, but branded drugs do, so this creates potential measurement error for branded drug prices. To address this, we rerun our demand estimation allowing branded drugs to be charging 30% rebates (following the findings in Kakani et al. (2020) and Kakani et al. (2022)), and obtain very similar results.

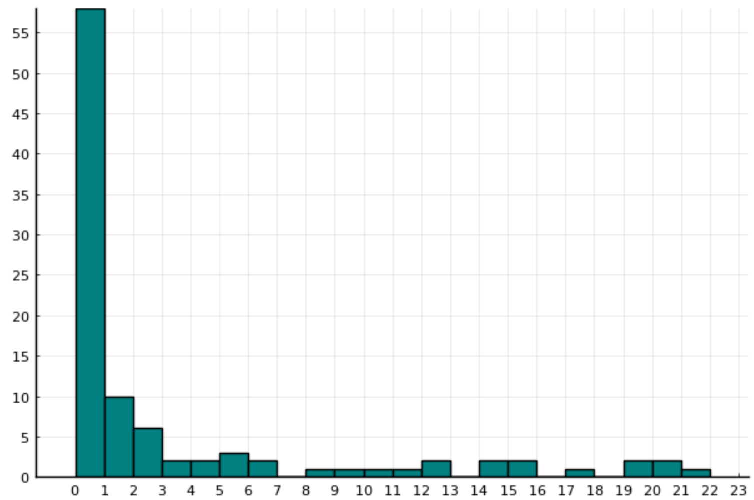
While rebates could be significant before loss of exclusivity, the size of the branded drug's rebate *after* loss of exclusivity is unknown. A lot of the theoretical and empirical literature on pharmaceuticals state that even after controlling for rebates, branded drug prices are significantly higher than generic prices. This fact is also used to explain the release of the AG as stated in FTC (2011).

[9]We searched news stories about every molecule online to try to ensure there were no non-economic events surrounding them that would affect generic entry. Molecules which faced unnatural events such as lawsuits being settled then revived, facing new lawsuits after entry and being forced to discontinue, facing public pressure such as Epipen, etc. are left out of the final sample.

[10]The FDA list on AGs can be found here: `https://www.fda.gov/media/77725/download`.
The online source by AG producers can be found here: `https://www.authorizedgenerics.com/product-finder/`
The FDA list on Paragraph IV lawsuits can be found here: `https://www.fda.gov/media/133240/download`.

[11]This raises the question of why we define a market at the molecule-formulation level instead of a higher level, such as the therapeutic class. First, we include a molecule-formulation fixed effect in the demand function to account for presence of other molecule-formulations in the same therapeutic class. Second, similar market definitions have been used in various antitrust proceedings and empirical papers (Conti and Berndt

Figure 2: Histogram of time-difference between first generic entry and AG release period.



This figure is limited to 22 quarters, while in the dataset 4 authorized generics get released later than 22 quarters. This distribution has a 10th percentile of 0, 25th percentile of 0, median of 0, 75th percentile of 4.25, and 90th percentile of 16.4. See the Appendix for the exact distribution.

3.2 Descriptive Statistics

While our dataset has comprehensive coverage across all prescription drugs sold in US, we look at a smaller subset of drugs for our project. We select molecule-formulations that saw its first generic entrant after 2004 and before 2016[12], for which we can identify presence or absence of AG with certainty, and those that are not surrounded by strong

(2020), Starc and Wollmann (2023), Wang et al. (2022), Yurukoglu et al. (2017)). Third, the fewer number of products allows us to capture substitution patterns between these bioequivalent products more accurately. Fourth, the branded drug's patent and generic approval holds at molecule-formulation level. Fifth, one drug can be in multiple therapeutic classes which makes modeling therapeutic classes as markets unclear. Finally, there is evidence suggesting that while there can be substitution across molecule-formulations within the same therapeutic class, usually the substitution is from a molecule-formulation without a generic to a molecule-formulation with one (Arcidiacono et al. (2013)). In this data we only look at periods after generic entry, so it is unlikely that people are switching away from the molecule-formulation we are studying.

[12] We now discuss how this sample selection ties in with our research question and interpretation of results. For studying AG entry, we believe this way of identifying markets AG may enter is quite realistic. AGs are always only released in response to generic entry. In fact, FTC (2011) show company documents where, when outsourcing AG manufacturing, brand manufacturer imposes strong restrictions on the partner AG manufacturers to prevent them from launching the AG before the first generic has entered. Therefore, identifying markets with at least one generic as markets where the AG may enter is realistic, and so our AG cost estimation should be unaffected.

Table 1: Descriptive statistics

	No. of molecule-formulations
Total	241
AG released	109
Formulation: Oral	162
Formulation: Injectable	58
Formulation: All Others	21
Faced Paragraph IV lawsuit	62

external circumstances (e.g. media outrage, serious or repeated lawsuits). This leaves us with 241 molecule-formulations. Of these 241 molecule-formulations, 109 see an AG released.

Table 1 shows descriptive statistics about our selected molecule-formulations: of the 241 molecule-formulations, about 45% see an authorized generic released, and over 67% are oral formulations. Note that each molecule-formulation has one branded drug and can have at most one AG.

For a molecule-formulation, we define the period of loss-of-exclusivity (LOE) to be the earliest quarter that saw positive generic drug sales.

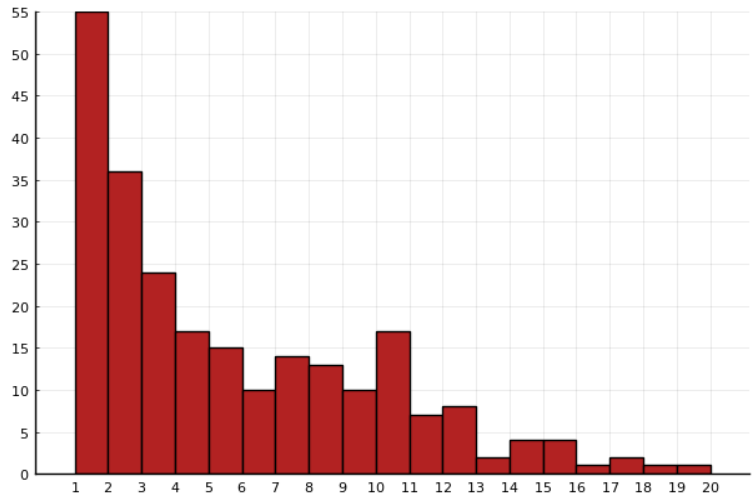
While Table 1 describes how many markets saw an AG released, Figure 2 shows when the AGs were released in these markets relative to generic entry. More specifically, Figure 2 is a histogram that shows how many quarters after first-generic-entry is AG released. 58 markets see AG release immediately with the first generic entry, 10 markets see AG release one quarter after the first generic entry, and 6 markets see AG released two quarters later. AGs continue being released sporadically afterh the third quarter, with the latest release happening 41 quarters after loss-of-exclusivity.

Next we look at the competitive landscape in our selected markets, specifically regarding generic presence. Figure 3 shows the histogram of total generics that enter into a

For studying generic entry, this means we rule out markets which are so unprofitable that a generic never enters. Most important prescription drugs see generic entry, so it is unclear how much we lose from not considering those markets.

One type of market we do rule out with this is a market where no generic enters, but would have entered had it not been for the threat of AG release. This shouldn't affect our demand estimation or AG cost estimation. For our per-period fixed costs we derive approximate bounds using markets with multiple generics and no AGs.

Figure 3: Histogram of total generic entry by molecule-formulation



Notes: This distribution has a 25th percentile of 2, median of 4, 75th percentile of 8, and 90th percentile of 11. There are 241 molecule-formulations.

molecule-formulation over time. This reveals that most markets have between 1-6 generic competitors.

4 Model

To study market dynamics after loss of exclusivity, we develop a structural model of entry and price competition between authorized generics and generics. The aim is to develop a stylized model that is tractable enough for estimation but captures the important economic mechanisms. On the demand side, we set up a discrete choice model of demand, where the consumer first chooses a group (brand, nonbrand, or outside option) then chooses a product within the group. On the supply side, we construct a two-stage model. The first stage is a static game of generic entry. The second stage is a dynamic game of Authorized Generic release decision. Every period, generics who decided to enter are randomly approved for product launch by the Food and Drug Administration (FDA), the branded drug manufacturer decides whether to release an AG, and the market participants compete in prices.

4.1 Demand

We model consumer demand as proceeding in two stages. In Stage 1, for a given molecule-formulation, each consumer chooses a group – brand, nonbrand, or outside option (forgoing medication entirely, switching to another molecule, etc.). The non-brand group includes both generics and AG (if present). In Stage 2, each consumer picks a product from the group chosen in Stage 1. Consumer choice is driven by the characteristics and prices of each product.

Stage 1: For molecule-formulation m, let g denote a group. Thus:

$$g \in \{\text{brand}, \text{non-brand}, \text{outside option}\}$$

In Stage 1, the Consuming Agent (a mix of the patient, provider, and insurer) chooses a group. The utility of consuming agent i for group g in market m at time t is given by:

$$u_{igmt} = \gamma_m^{(1)} + \lambda_t^{(1)} + \alpha_i^{(1)} ln\, p_{gt} + \beta_{1i}^{(1)} \text{brand}_g + \beta_2^{(1)} brand_g \cdot \text{time-since-loe}_t + \xi_{gt}^{(1)} + \varepsilon_{igt} \tag{1}$$

where $ln\, p_{gt}$ denotes the average of log prices within the group, "brand" is an indicator for the brand group, and $\xi_{gt}^{(1)}$ allows for unobserved group-specific quality. The variable "time-since-loe" measures the number of quarters since generic entry (loss of exclusivity), and this is used to form a brand-specific time trend.[13] The random coefficients $\alpha_i^{(1)}$ and $\beta_i^{(1)}$ allow for heterogeneity in price sensitivity and brand valuation respectively. As is common in the empirical literature, we further assume $\alpha_i^{(1)} \sim \mathcal{N}(\alpha^{(1)}, \sigma_\alpha^2)$ and $\beta_i^{(1)} \sim \mathcal{N}(\beta_1^{(1)}, \sigma_1^2)$.

Note that the brand group has a single product, while the non-brand group can have multiple products. Thus $ln\, p_{gt}$ is the branded drug's price for the brand group, and is the *mean* of all non-branded drugs' prices for the non-brand group. The price of the outside option is normalized to zero as is standard in the discrete-choice estimation literature.

Stage 2: If the Consuming Agent chose the non-brand group in Stage 1, and there

[13] Branded drug manufacturers can sometimes take actions that slow down generic adoption. This involves things like product hopping, getting approval for new therapeutic uses, gaining orphan drug designation, etc. These moves stop consumers and pharmacists from switching over to generic drugs and get around automatic substitution laws. While our model does not specifically account for such actions, our demand specification contains a brand-specific time trend that aims to capture the idea that the branded drug loses its appeal over time as the manufacturer runs out of such tactics. As would be expected, we find that the branded drug becomes less attractive over time, possibly because such tactics become less effective over time.

are multiple non-brand drugs available for the molecule formulation m at time t, then the patient has to choose from among these products.

In Stage 2, the Distributing Agent (a mix of the pharmacy, procurement officer at the provider, and other intermediaries) help the patient choose a specific non-brand product. The utility of Distributing Agent i choosing non-brand product j in market m at time t is given by:

$$u_{ijt} \;=\; \gamma^{(2)}_{m(j)} + \alpha^{(2)} ln\, p_{jt} + \beta^{(2)}_1 \mathrm{AG}_j + \xi^{(2)}_{jt} + \varepsilon_{ijt} \tag{2}$$

where $ln\, p_{jt}$ denotes the log price of product j, AG_j is a dummy for product j being an Authorized Generic, and $\xi^{(2)}_{jt}$ allows for unobserved product-specific quality.

The market share of group g is given by integrating over the distribution of random coefficients:

$$s_g = \int \frac{e^{\delta^{(1)}_g + \mu_{ig}}}{\sum_l e^{\delta^{(1)}_l + \mu_{il}}} f(\mu_i|\tilde{\theta}_2) d\mu_i \tag{3}$$

where $\theta_2 = (\sigma_\alpha, \sigma_\beta)$, $\delta^{(1)}_g = \gamma^{(1)}_m + \lambda^{(1)}_t + \alpha^{(1)} ln\, p_g + \beta^{(1)}_1 \mathrm{brand}_g + \beta^{(1)}_2 brand_g \cdot \text{time-since-loe} + \xi^{(1)}_g$, and $\mu_{ig} = \sigma_\alpha \nu_{\alpha,i} ln\, p_g + \sigma_\beta \nu_{\beta,i} \mathrm{brand}_g$.

Conditional on choosing nonbrand group, the probability of a consumer choosing product j from nonbrand group is given by:

$$s_{j|g} = \frac{e^{\delta^{(2)}_j}}{\sum_k e^{\delta^{(2)}_k}} \tag{4}$$

where $\delta^{(2)}_k = \gamma^{(2)}_{m(k)} + \beta^{(2)} \mathrm{AG}_k + \alpha^{(2)} ln(p_k) + \xi^{(2)}_k$.

Stage 1 gives the expression for the market share of a group, while Stage 2 gives the expression for the market share of a product conditional on a group. Therefore, we can express the market share of a single good in terms of group market share and market share of good conditional on choice of group:

$$s_j = s_{g(j)} s_{j|g(j)} \tag{5}$$

See the Appendix B for a detailed derivation and discussion of Equation 5. Therefore, estimating the parameters in Stage 1 allows us to predict $s_{g(j)}$, and estimating the parameters in Stage 2 allows us to predict $s_{j|g(j)}$. This allows us to study counterfactual scenarios and how they affect market shares of all products. Futhermore, estimating this demand system amounts to estimtaing a random-coefficient discrete choice model over groups, then a multinomial logit model over the nonbrand products.

4.1.1 Discussion on demand model

We now give a discussion of our modeling choice for the demand system, specifically why we choose the two-stage demand setup instead of other demand setups. Generally, papers estimating demand systems follow the setup of Berry et al. (1995), where each consumer chooses from a set of products. When trying to account for correlated preferences within a group, some papers use a random-coefficients or nested logit setup . In contrast, in our setting, a consumer chooses a group, then chooses a product within a group. The reasoning for this is that we found, after experimenting with different specifications, that this matched the empirical patterns of prices and market share best. In our data we see that the price difference between the brand and non-brand group leads to substantial switching from branded drug to non-branded drugs, which implies high price elasticity of consumers. However, within non-brand products, we see very little switching based on specific prices of non-brand drugs. That is, a higher-priced generic has only slightly lower market share than a lower-priced generic. This then suggests consumers are not price elastic. When all products are estimated together in a single discrete choice setting like Berry et al. (1995), we end up finding very low price-elasticity overall for the market. When we do counterfactuals, this low price-elasticity leads to unrealistic predictions such as low substitution between brand and generics after loss-of-exclusivity (which is at odds with what we see in the data).

Our specification gets around this by capturing the right substitution patterns in the data. Stage 1 captures the fact that the price gap between brand and nonbrand leads to a large substitution towards the cheaper nonbrand products. Stage 2 picks up the mild price elasticity between generics, while also helping explain why average prices decline as more generics enter the market. While our demand model is somewhat different from other discete-choice settings, note that our aim for the project is to study entry decisions; as long as we correctly predict prices and per-period profits, our central arguments work.

4.2 Supply

We first present an overview of our supply model. The following subsections explain the components of the model, and Section 4.3 goes over the reasoning behind the modeling choices.

Recall that generic manufacturers have to wait for FDA approval before they launch for approval; as a result, they enter (apply for ANDA approval) well in advance. In contrast, authorized generics can be released at any time. This motivates our model setup below.

There are two stages to the supply side:

1. In the first stage, generic firms decide whether to enter a market (i.e. apply for ANDA to molecule-formulation) or not. This stage is modeled as a game of simultaneous entry between identical firms, i.e. firms apply for ANDA simultaneously until the value of entering no longer exceeds the entry cost.[14]

2. In the second stage, loss of exclusivity happens and a dynamic game begins. The dynamic game lasts T periods. Every period, a random number of the generic firms which applied for ANDA are approved and introduced into the market. The branded drug manufacturer decides whether to release AG or not (where the AG release decision is irreversible). Then, all the firms set prices and receive payoffs.

We now explain the model in reverse, starting from payoffs in the second stage. Throughout we suppress the market subscript m unless necessary for exposition.

4.2.1 Per-period payoffs

The branded drug manufacturer's per-period payoff is:

$$\pi^b(s_{mt}) = [P^b_{mt} - MC^b_m] s_b(s_{mt}) M_m - \phi_{mb} + \mathbf{1}(AG_{mt} = 1) \left[[P^{AG}_{mt} - MC^{AG}_m] s_{AG}(s_{mt}) M_m - \phi_{mAG} \right] \tag{6}$$

where s_{mt} denotes the state variable vector in market m, superscripts and subscripts b and AG refer to the branded product and AG respectively, P denotes price, MC denotes marginal

[14] An alternative way of doing this would be to assume sequential entry, i.e. firms apply for ANDA sequentially until the value of entering no longer exceeds cost. Under our assumption of identical generics, both simultaneous and sequential entry will yield the same number of generic entrants.

cost, $s(s_{mt})$ denotes market share, ϕ_m is the per-period fixed cost in market m, and M_m denotes market size. The indicator variable $\mathbf{1}(AG_{mt}=1)$ denotes whether an AG has been released onto the market. The intuition behind the branded drug manufacturer's payoff is that it makes the first half of the payoff from the branded drug, and the second half from the AG conditional on releasing the AG.

Similarly, the generic firm l's per-period payoff with the firm-specific state variable vector $s_{l,t}$ is:

$$\pi^g(s_{lmt}) = (P^g_{mt} - MC^g_m)s_g(s_{lmt})M_m - \phi_{mg} \tag{7}$$

The branded drug, generics and AG compete on prices. We use a reduced-form price prediction regression to predict prices in different states and markets; see Section 5.1. While the model has been written for maximum generality, during estimation we set $MC_b = MC_{AG} = MC_g$ for a given market m. This assumption can be relaxed.

4.2.2 Second-stage

A dynamic game begins from $t = 1$ and lasts T periods, where every period is a quarter. Let n^*_e be the number of generic firms that have applied for an ANDA (which is determined in the first stage). In period $t = 2$ the branded drug's market exclusivity expires, and every period a random number of generic firms gain FDA approval and enter the market.

Since AG release is a dynamic decision, the branded drug manufacturer's choice problem can be written in terms of value functions. Thus, the value function for a branded drug manufacturer every period is given by:

$$\begin{aligned} V^b(s_{mt},\varepsilon_{mt}) = \max_{AG_{mt+1}\in\{0,1\}} \pi^b(s_{mt}) - \mathbf{1}(AG_{mt}=0, AG_{mt+1}=1)\kappa^{AG}_m + \\ \beta E[V^b(s_{mt+1},\varepsilon_{mt+1})|s_{mt},\varepsilon_{mt}] + \varepsilon_{mt}(AG_{mt+1}) \end{aligned} \tag{8}$$

where $AG_{mt+1} = 1$ means the AG is in the market m during period $t+1$. AG entry cost in molecule-formulation m is denoted as κ^{AG}_m. The AG entry decision is irreversible, i.e. $AG_{mt} = 1$ implies $AG_{mt+k} = 1\ \forall k > 0$. AG entry is implemented with a one-period delay.

The timing of AG release could be influenced by factors observed to the manufacturer but unobserved to the econometrician. To account for such factors, the model (following the existing literature on dynamic games) includes choice-specific payoff shocks $\varepsilon_{mt}(AG_{mt+1})$.

These shocks follow a Type 1 Extreme Value distribution. Henceforth, we refer to these choice-specific shocks as "AG timing shocks", since they primarily affect when the AG gets released in the market. A more detailed discussion is given in Section 4.3.

Similarly, the value function for generic l is given by:

$$V^g(s_{lmt}) = \pi^g(s_{lmt}) + \beta E[V^g(s_{lmt+1})|s_{lmt}] \tag{9}$$

where $s_{l,t}$ includes whether generic l has been approved for production by the FDA. While the generic manufacturers do not face a dynamic decision in the second stage, they use the value function to make entry decision in the first stage.

After period T, the market state is set at s_{mT}, and the manufacturer receives the corresponding payoff for infinite periods:

$$V^b(s_{mT}) = \sum_{\tau=T}^{\infty} \beta^\tau \pi^b(s_{mT}) \tag{10}$$

Similarly for generics:

$$V^g(s_{mT}) = \sum_{\tau=T}^{\infty} \beta^\tau \pi^g(s_{mT}) \tag{11}$$

The manufacturers have to form expectations over generics' approval probabilities; recall that approval is given randomly to entrants over time. To express the value functions, we need to pin down how state transitions happen regarding generic approval. Thus, we model the FDA approval rate as a binomial process. This follows Ching (2010).

Suppose by period t in market m, $\mathcal{N}_{mt}$ generic entrants have already received approval and launched their products. Thus $n^*_{em} - \mathcal{N}_{mt}$ entrants have not yet received approval . The probability that of the $n^*_{em} - \mathcal{N}_{mt}$ remaining entrants, k will receive approval in this period is given by:

$$P_e(k, \mathcal{N}_{mt}, t) = \binom{n^*_{em} - \mathcal{N}_{mt}}{k} \lambda(t)^k (1-\lambda(t))^{n^*_{em} - \mathcal{N}_{mt} - k} \tag{12}$$

where $\lambda(t)$ – the probability of a single generic receiving approval – is estimated from the data. We assume the branded drug manufacturer knows n^*_e and total generics that received approval for $t = 2$ ($\mathcal{N}_{m2}$).

For a molecule-formulation m, the incumbent brand manufacturer draws the AG entry cost κ^{AG}_m from the following distribution:

$$\kappa_m^{AG} = \begin{cases} e^{\bar{\kappa}}, & \text{w.p. } p \quad \text{“trade shock”} \\ \infty, & \text{w.p. } 1-p \quad \text{“no-trade shock”} \end{cases} \tag{13}$$

That is, with probability p releasing an AG costs $\bar{\kappa}$, and with probability $1-p$, releasing an AG is prohibitively expensive. For simplicity we refer to the former as the AG drawing a “trade shock” and the latter as a “no-trade shock” from the entry cost distribution. In Section 4.3 we discuss this modeling choice.

4.2.3 First-stage

In the first stage, an infinite number of generics decide whether they want to enter. We assume all generic firms are ex-ante identical, do not receive private error draws for entering and staying out, and do not know their draws of ξ_{jt} conditional on entry.

Generics keep entering until the value from entering no longer exceeds the cost of entry. More precisely, the equilibrium number of generic entrants is n_e^*, determined by the following inequality:

$$V^g(s_{gm0}, n_{em}^*) \geq \kappa_m^g > V^g(s_{gm0}, n_{em}^* + 1) \tag{14}$$

where κ_m^g denotes the generic manufacturer's entry cost for molecule-formulation m and s_{gm0} is the initial state of the market from a generic's perspective.. Intuitively, the value function of entry is declining in the number of entrants - more entrants mean more competition in the market and lower per-period profits. At the equilibrium n_e^*, current entrants can cover their entry cost, but an additional entrant would stop that from happening. This partly follows Igami (2018). Finally, unlike the branded drug manufacturer, the generics don't know $\mathscr{N}_{m2}$.

4.2.4 Model solution

We solve the model by backward induction and recover the equilibrium number of generic entrants and conditional choice probabilities (CCPs) of authorized generics. Since the AG's CCPs enter the value functions of generic entrants and the AG knows n^*, all agents are forming rational expectations about each others' choices when making a decision. This means that a policy or environment change that affects the choice probabilities of any agent will have feedback effect on other agents in the market. The details of the model solution are given in Appendix A.1.

4.3 Discussion

Economic interpretation of AG timing shocks and their variance: Recall that the authorized generic release decision faces "AG timing shocks", which are choice-specific payoff shocks $\varepsilon_t(AG_{t+1})$ that follow a Type 1 Extreme Value distribution. These logit shocks have a logit scaling parameter σ_ε that governs the variance of these shocks. The higher the σ_ε, the greater the variance of choice-specific logit shocks, so the greater the authorized generic release decision will deviate randomly from the optimal release decision. Intuitively, this is used to account for payoff shocks observed to the agent but unobserved to the econometrician. During estimation, it is used to reconcile why we observe firms in similar states of the world make different choices.[15]

In our model, the logit scaling parameter chiefly explains why authorized generic launch is delayed by some brand manufacturers far past the loss-of-exclusivity period. In general, the most profitable quarters for a non-brand product are those right after loss-of-exclusivity, because there are few competitors. Over time, more generics gain FDA approval and enter, moving prices closer and closer to marginal cost. While authorized generic release also accounts for the knock-on effect on branded drug revenue, it is generally optimal for the authorized generic to be released early; yet, in the data, we see that a large fraction of authorized generics get released several quarters after loss-of-exclusivity. The logit scaling parameter rationalizes this behavior by ascribing the delay to unobserved payoff shocks not observed to the econometrician.[16]

So what could be plausible explanations behind such delays or unobserved payoff shocks? First, this could be driven by manufacturing delays in setting up production lines. Second, branded manufacturers may outsource authorized generic production to other firms

[15] This is a common methodology used in dynamic discrete choice models with dollar-valued payoffs. See Igami (2018) and Alam (2023). Papers which have unitless payoff functions instead normalize the logit scaling parameter to 1.0 and estimate all other structural parameters relative to it, as in Rust (1987).

[16] To be exact, the logit shocks also help explain why AG enters in unprofitable markets, as well as why an AG enters later when it should enter earlier. However, in our setting, we find that the logit shocks chiefly serve to delay entry into markets where the AG is profitable.

(such as Prasco and Greenstone), and there could be contracting frictions and bargaining between these parties. Third, there could be organizational friction within the branded manufacturer that slows down the decision-making process. Fourth, it could be that the loss-of-exclusivity was unanticipated or there was uncertainty surrounding it, which meant that the branded manufacturer was unprepared for the loss-of-exclusivity when it happens. Fifth, there could be policy uncertainty such as how the FDA would approve or regulate the new generic entrants, or how a lawsuit might be resolved, that may add further uncertainty to the release decision. Finally, it could be that the branded manufacturer is engaging in other generic-deterring strategies (such as product-line extensions) which affects its AG timing decision.

Motivation behind no-trade shock: We model authorized generic's entry cost as a bimodal distribution, i.e. a "trade shock" and a "no-trade shock".[17]

For our estimation, we find that this setup yields more reasonable estimates; it helps explain why a few seemingly-profitable markets are not entered by AGs. From an economic point of view, the "no-trade shock" tries to subsume a variety of factors (unobserved to us) specific to the manufacturer or the market that makes it untenable to release an authorized generic. There could be several possibilities. First, the branded drug manufacturer may not have any generic subsidiaries. This makes it very difficult to release an authorized generic, because marketing a non-branded drug versus a branded drug entail very different skillsets.[18] Second, the brand manufacturer may choose to withhold an authorized generic due to a no-authorized-generic-settlement.[19] Unfortunately we do not have

[17] We have also done some preliminary work with a lognormal distribution instead of bimodal distribution, and found similar results.

[18] The marketing of a branded drug focuses on advertising and detailing, negotiating formulary positions by insurer, etc. The marketing of a generic drug focuses on extreme cost efficiency, winning pharmacy auctions (Cuddy (2020)), etc. These are very different skillsets, and for a branded drug, it may not be worth the hassle of getting into the generics business just to release an authorized generic. It could instead outsource its production to another generic firm (for instance, Prasco and Greenstone specialize in bringing authorized generics for other firms into the market) but that involves negotiation cost and profit-sharing.

[19] "No-AG settlements" are settlements where the generic firm agrees to not enter immediately in exchange for assurance that an Authorized Generic will not be released. Such a no-authorized-generic-settlement could be related to a Paragraph IV settlement regarding the same molecule-formulation, or a settlement involving a different molecule-formulation but involving the same firms. This paper does not model No-AG settlements explicitly, partly because it is extremely difficult to identify which Paragraph IV lawsuits resulted in such settlements. Our demand model is not affected by No-AG settlements, the no-trade shock for AG subsumes No-AG settlements, and our counterfactuals apply to markets where such settlements have not occurred. We include a brief discussion on No-AG settlements through the lens of our model in Section 7.4. In addition,

data on whether a no-authorized-generic-settlement deal was reached, and so it would be subsumed by the "no-trade shock" in our model. Finally, it could be that the brand is engaging in other generic-deterring strategies such as product-line extensions which limits its ability to release an Authorized Generic due to resource constraints.

Per-period fixed costs: We estimate approximate values of the per-period fixed costs. Per-period fixed cost reflects the cost of operating the factory and production line, maintaining supply chains, transaction costs, etc. It is also the economic cost of marketing a product, and so reflects costs beyond accounting costs. For instance, there could be opportunity cost tied to allocating inputs to this product as opposed to elsewhere, or there could be additional constraints facing the managers.

For estimation purposes, we find that allowing for per-period fixed costs achieves two things. First, it rationalizes generic entry numbers for publicly reported generic entry costs (which are between $3.8 million and $15 million; see Starc and Wollmann (2023)). Without any per-period fixed cost, at these reported generic entry costs, the model predicts very high generic entry numbers. This isn't just a feature of the model; looking at the lifetime revenues generated by generics in our data, they vastly exceed the reported generic entry costs, which implies there are missing costs that explain why more generics do not enter. Second, we find that it gives reasonable numbers for AG entry costs. Without it, AGs are predicted to make very large profits, which has to be rationalized by high values of $ka\bar{p}pa$. However, intuitively $ka\bar{p}pa$ should be lower than generic entry costs, because the AG does not have to reverse-engineer the drug like generic manufacturers. In short, the per-period fixed costs scale down per-period profits and help us better match the data and reported costs.[20]

Discussion on price prediction equation: Instead of assuming a pricing conduct such

see FTC (2011) for a more detailed analysis.

[20]One simplification we make is that we do not model exit, which means firms don't exit even when per-period economic profit is negative. If shutting down prevents the manufacturer from incurring the per-period fixed costs, then manufacturers may shut down operations when economic profit is negative due to the presence of per-period fixed costs. We make this simplification to keep the model solution tractable - exit is a dynamic decision, so allowing for exit would require us to model a dynamic game of exit among all firms in the second stage. The decision to exit comes with a separate interesting set of economic motives such as war of attrition (see Takahashi AER 2014). One possibility is to have a myopic exit-decision rule where generics with negative per-period profits randomly shut down for a period until per-period profits of all remaining firms is weakly positive. When we redo our counterfactuals with no per-period fixed costs, the qualitative results of our counterfactuals are very similar.

as Nash-Bertrand or Nash-in-Nash bargaining, we use a price prediction regression to predict prices charged by different firms in different markets. We now discuss our reasoning for this approach and the trade-offs involved.

One pricing conduct assumption we could have made is Nash-Bertrand pricing, which however implies that all bargaining power is upstream with the pharmaceutical manufacturer. This is unlikely to be true in reality, since there are many intermediaries - Pharmacy Benefit Managers (PBMs), wholesalers, pharmacies, providers, etc. - that negotiate drug prices with substantial bargaining leverage. This is also borne up by the estimated price elasticities in our paper - we find most price elasticities to be less than 1, which aligns with a model of split bargaining power between upstream and downstream firms rather than a take-it-or-leave-it offer (TIOLI) from upstream firms. Since we have aggregate data instead of detailed plan-level or PBM-level data, we cannot explicity model the pricing game between manufacturers and all the intermediaries.[21] The advantage is that our predicted per-period profits are not susceptible to misspecifications about the pricing conduct assumption.[22]

Note that in our counterfactuals, we do not change any primitives that would affect the static pricing game. Otherwise, the price prediction equation estimated from the observed

[21]Other papers which study pharmaceuticals with aggregate data and make a pricing conduct assumption include Arcidiacono et al. (2013), Dubois et al. (2022), and Starc and Wollmann (2023). Unlike these papers there are major differences in our analysis that precludes us from doing the same. Arcidiacono et al. (2013) focus on a small number of molecules, which allows them to leverage additional data and impute rebates. In contrast, our paper considers well over 200 molecule-formulations. Dubois et al. (2022) restrict their attention to sales in the hospital channel for drug sales, while we consider all channels. They assume Nash-Bertrand, which is a more reasonable assumption in their setting because hospitals are not covered by large intermediaries such as PBMs, and price elasticities exhibited in these non-patient-facing channels are much larger. Starc and Wollmann (2023) consider the pharmacy channel and use different datasets compared to this paper.

[22]A regression of prices on level of competition may suffer from endogeneity issues from demand shocks. This is less of an issue than in other settings, because generics cannot simply respond to a demand shock and enter the market - they need to wait years to get FDA approval for marketing. Furthermore, we also add in molecule and formulation fixed effects.

We also cannot impute marginal cost rigorously from observed prices, as is done in Dubois et al. (2022) and Starc and Wollmann (2023). (Recall that we set the marginal cost of a molecule-formulation to be the minimum observed price for that molecule-formulation in our dataset.) However, if our pricing conduct assumption is incorrect, the imputed marginal costs would have been incorrect as well. Other approaches by authors who do not impute marginal cost by assuming a pricing conduct such as Nash-Bertrand include Tunçel (2024) (who uses the minimum observed price to construct moment inequalities about non-negative costs) and Grennan et al. (2021) (who set marginal cost as 0.17 times generic price).

data would no longer be a good predictor, as a change in demand primitives should change the underlying pricing patterns. Instead, we limit our counterfactuals to changing parameters that affect dynamic incentives (such as entry restrictions or faster approval rates).

Our price prediction equation is also similar in spirit to the price-setting equation in Maini and Pammolli (2023). In their paper, the authors try to predict prices across different European countries with different bargaining policies and unknown objective functions. Instead of taking a strong stand on either, the authors opt for an agnostic approach to predicting prices and use a flexible control function. Our motivation is similar in that we want to be agnostic about how different intermediaries interact across the supply chain. In both our paper and Maini and Pammolli (2023), the main focus is on entry instead of the static pricing game, which means being able to accurately predict per-period payoffs is sufficient for our analysis.

Another issue is that during this period, some generic manufacturers may have been engaging in price-fixing (Cuddy (2020), Starc and Wollmann (2023)). This occurred from late 2013 onwards; our dataset's coverage is from 2004-2016. As a result, we rerun the price prediction regression by dropping years 2013 onwards, and obtain very similar results.

Generic entry as a static entry game: Recall that we model the entry decision by generics to be a static entry game that happens before loss-of-exclusivity (following Ching (2010)). We choose to model generic entry as a static game instead of a dynamic entry game for the following reasons. In the pharmaceuticals industry during our sample period of 2004-2016, the mean approval time for ANDA was about 40 months, so any decision to enter was implemented with significant delay. As a result, generic manufacturers needed to file for ANDA well in advance of patent expiration. Note that the AG does not need FDA approval as it is riding on the branded drug's original approval. As a result, we model AG entry as a dynamic decision being made every period.

Another simplification we make is that while generics apply well in advance of patent expiration, they do not know exactly when they will get FDA approval. In reality there are sometimes cases where a generic firm applies for and gains ANDA approval well in advance of patent expiration, and so can start production in the same quarter when patent expires. Our estimation of generic launch rates partly accounts for that; see Section 5.1 for a detailed explanation.

Information assumptions: To simplify computation, we make two assumptions on information regarding the number of generic entrants. First, we assume that the AG knows how many generics have filed for ANDA to its molecule-formulation. In reality, the information on ANDA application by a generic manufacturer becomes public only after it wins ANDA approval from the FDA (although generic manufacuterers may have private information sources that could alert them to ANDA filings by rivals). This is a simplifying assumption that improves the tractability of our model. Second, we assume that the authorized generic correctly predicts the number of generic launches in the quarter when loss-of-exclusivity happens; this is realistic as the FDA would have announced before loss-of-exclusivity which ANDAs have been approved by then. Third, we assume that generics know the value of κ_m^{AG}. This is in keeping with the IO literature, which assumes that rivals know each others' cost structures; this assumption can however be relaxed in our model.

5 Estimation

We use a reduced-form price regression to predict prices, and estimate generic approval rates from the data. Demand is estimated using methods from Berry et al. (1995) and Maggio et al. (2022). The supply model is solved via backward induction, and AG cost parameters are estimated via simulated method of moments. We end with a heuristic discussion on identification.

5.1 Pricing and generic approval rates

Instead of assuming a pricing conduct (such as Nash-Bertrand or Nash-in-Nash bargaining), we choose to be agnostic about the price-setting behavior and instead predict prices using market-specific observables. That is, we regress observed prices on covariates such as molecule fixed effects, formulation fixed effects, number of generics, presence of branded drugs, etc. This regression is then used to predict prices in different states of the world. See Section 4.3 for a detailed discussion on why we do this instead of making a pricing conduct assumption such as Nash-Bertrand. The price prediction regression gives expected results - nonbrands charge lower prices as more nonbrand products enter, brands raise prices when facing generic entry, brands charge higher prices than nonbrands, and injectables are more expensive than oral formulation drugs.

Table 2: Price prediction regressions

	Patient-facing	Non-patient-facing
formulation: Injectable	2.118	–
	(0.136)	(–)
formulation: Oral	-1.308	–
	(0.112)	(–)
No. of nonbrand drugs	-0.269	-0.298
	(0.009)	(0.016)
No. of nonbrand drugs squared	0.008	0.010
	(4.843e-04)	(0.001)
Brand	0.123	-0.076
	(0.031)	(0.041)
Brand * time-since-generic-entry	0.017	-0.005
	(0.001)	(0.002)
Brand * No. of nonbrand drugs	0.232	0.231
	(0.004)	(0.008)
Brand * Authorized generic present	0.041	0.774
	(0.031)	(0.064)

Notes: We regress log of prices on the covariates listed above. Robust standard errors are in parentheses.

We make several simplifying assumptions. First, for a molecule-formulation, we set the marginal cost equal to 0.95 times the minimum observed price in our data. Second, we assume that all non-brand drugs charge the same price in a market-quarter[23] Third, we leave out quarter fixed effects in our demand system, and instead add the median estimated quarter fixed effect to the demand system's intercept term.

We combine this price prediction equation and marginal costs with our estimated market share equation (Equation (5)) to generate per-period profits in different states of the world.

To operationalize Equation (12) we also estimate the per-quarter generic approval rates $\lambda(t)$, where t denotes quarters since loss-of-exclusivity (and not calendar time). We estimate the average generic launch rates at every quarter since loss-of-exclusivity, then use a best-fit line through the estimated rates. These predicted rates are used in our main estima-

[23] In our data we see AGs price close to other generic manufacturers in the market.

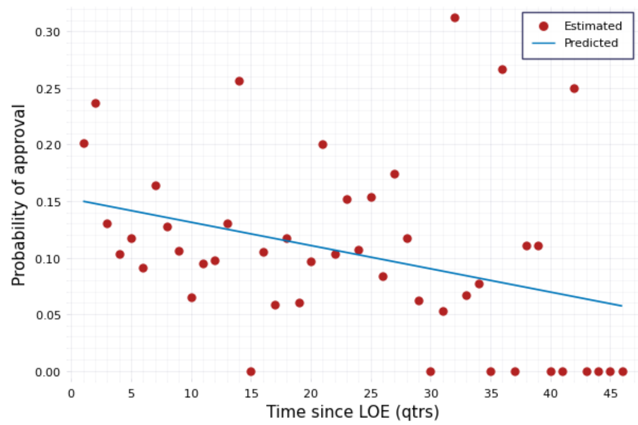


Figure 4: Generic entry rate

tion.[24]

Note that the estimated generic launch rates comprise not just the FDA's approval rate but also generics applying early in an effort to get FDA approval close to the loss-of-exclusivity. In particular, some generics apply early so that they can be in the market as soon as loss-of-exclusivity happens, while others try to be there at least soon after loss-of-exclusivity. We estimate generic launch rates to be highest at the beginning of loss-of-exclusivity, and declining over time; this reflects the idea of anticipating loss-of-exclusivity described above. As the FDA speeds up its approval rates, it should lead to the estimated entry rates shifting up for every quarter after loss-of-exclusivity.

5.2 Demand estimation

We estimate each stage of the demand model separately. Stage 1 of the demand model is a standard random-coefficients discrete choice model, so we use the method of Berry et al. (1995) (see Conlon and Gortmaker (2020) and Nevo (2000) for detailed discussions

[24]This is done as follows. For each molecule-formulation and quarter since loss-of-exclusivity, we calculate the number of generics waiting to enter, i.e. final number of generics observed in that molecule-formulation minus the current number of generics present in the molecule-formulation. We sum this number across all molecule-formulations at a given quarter since LOE to get the remaining entrants for every quarter since loss-of-exclusivity in our dataset. Then we find the net generic entry that happens in every each molecule-formulation and quarter since loss-of-exclusivity. We sum this number across all molecule-formulations at a given quarter since LOE to get the total net generic entry for every quarter since loss-of-exclusivity in our dataset. Taking the ratio of total net generic entry by remaining entrants at every quarter since loss-of-exclusivity gives us the average generic launch rates at every quarter since loss-of-exclusivity.

on implementation). We use Gandhi-Houde instrumental variables (Gandhi and Houde (2019)) and employ two-step GMM.

Stage 2 of the demand model is a multinomial logit demand model but without an outside option. Given that idiosyncratic errors follow Type-1 Extreme Value distribution, we can write market share of generic drug j conditional on nonbrand group being chosen in Stage 1, as:

$$s_{jt|g} = \frac{e^{\delta_{jt}^{(2)}}}{\sum_k e^{\delta_{kt}^{(2)}}}$$

where $\delta_{kt}^{(2)} = \gamma_{m(k)}^{(2)} + \beta^{(2)}\mathrm{AG}_k + \alpha^{(2)} ln(p_{kt}) + \xi_{kt}^{(2)}$. Using the transformation from Berry (1994):

$$ln(s_{jt|g}) = -ln(\sum_k e^{\delta_{kt}^{(2)}}) + \delta_{jt}^{(2)}$$

Plugging in the formula for $\delta_{jt}^{(2)}$ gives us the estimating equation:

$$ln(s_{jt|g}) = -ln(\sum_k e^{\delta_{kt}^{(2)}}) + \gamma_{m(j)}^{(2)} + \beta^{(2)}\mathrm{AG}_j + \alpha_2 ln(p_{jt}) + \xi_{jt}^{(2)} \tag{15}$$

Following Maggio et al. (2022) this can be estimated using a linear regression of logs of market share on market-by-quarter fixed effects and other covariates. That is, the first term on the right hand side varies at the market-by-quarter level, so including market-by-quarter fixed effects absorbs it. The remaining coefficients can just be estimated through a linear regression with fixed effects ($\xi_{jt}^{(2)}$ becomes the econometric error in the regression).[25]

We add an additional source of heterogeneity in our demand estimation. Our dataset contains a breakdown of drug sales by the channels were the sale happened. These channels are clinics, drugstores, federal facilities, food-stores, long term care hospital, HMO, home health care, mail services, and non-federal facilities (Berndt et al. (2017), Dubois and Majewska (2022)). We take advantage of this by allowing our demand parameters to

[25]By contrast, in Berry (1994) the authors use the outside good's market share to net out the term $ln(\sum_k e^{\delta_{kt}})$. In our case this cannot be done because there is no outside option in Stage 2; conditional on choosing the non-brand group in Stage 1, the consumer has to choose from one of the non-brand products in Stage 2.

coarsely vary based on the channel presence of a molecule-formulation, in the following manner.

We classify all molecule-formulation into one of two types: patient-facing or non-patient-facing. "Patient-facing channels" are channels where the individual patient has more influence over group choice, versus other settings. For instance, a patient buying from a pharmacy is allowed to specifically purchase a nonbrand group. However, a patient being administered medication at a hospital has much less influence over group choice. The channels we classify as "patient-facing" are retail pharmacies, mail pharmacies, and food stores. We can expect demand parameters to vary between these two categories because of the intermediaries and bargaining processes involved. In patient-facing channels, the insurer, patient, physician, and PBM together determine the product choice and prices; by contrast, non-patient-facing channels involve the provider negotiating with the wholesaler or manufacturer and determining product choice.

Note that generally most molecule-formulations get sold in most channels, and this categorization is done using the fraction of sales in either type of channels. That is, a molecule-formulation is classified as patient-facing if more than 50% of its sales are in patient-facing channels. In our data, we classify 183 molecule-formulations as patient-facing and 58 molecule-formulations as non-patient-facing.

5.3 Supply estimation

We estimate the supply-side cost parameters of our model. First we derive approximate bounds on the per-period fixed costs by matching predicted generic entry numbers to observed numbers at publicly reported generic entry costs. Next, we estimate the AG cost distribution and logit scaling parameter of the AG timing shocks. Throughout we set the quarterly discount factor at $\beta = 0.95$.[26]

[26]This lines up with an annual discount rates of 0.81, which has been used in prior empirical work such as Igami and Sugaya (2022). Some dynamics papers use higher annual discount rates such as 0.95. We choose a lower number because i) we do not account for exit or discontinuation of manufacturers, so the low discount rate may indirectly account for that possibility by downweighting future payoffs more, ii) pharmaceutical industry sees new molecule-formulations experiencing loss-of-exclusivity regularly so there's a high risk of a product market becoming obsolete in the future. Such risks of product obsolescence is implicitly captured a lower discount rate.

5.3.1 Estimating the per-period fixed costs

We derive approximate bounds on the per-period fixed costs. We isolate molecule-formulations for which AGs do not enter, and assume that AGs drew a no-trade shock for these molecules. For a given generic entry cost κ_g, we try different trial values of per-period fixed costs and solve for the equilibrium generic entrants in each market. Using grid-search, we pick the per-period fixed cost that minimizes the sum of squared differences between the observed and predicted generic entry numbers. Recall that reported genreic entry costs vary between \$3 million and \$15 million; therefore we set κ_g to each of these bounds and recalculate the per-period fixed costs. We find that per-period fixed costs range from \$400,000 to around \$2 million. Thus, we fix per-period fixed costs at \$1.2 million for our main estimation. Details of the estimation procedure are given in Appendix A.3.

We recognize that generic entry costs and per-period fixed costs may vary by molecule-formulation. Our aim is to have per-period fixed costs that scale down profits so that at the publicly-reported generic entry costs, the model predicts reasonable generic entry numbers and AG entry costs. While we likely obtain a noisy approximation of the per-period fixed cost through our procedure, the resulting estimate achieves our purpose.

5.3.2 Estimating the entry cost distribution of AGs

Next we estimate the logit scaling parameter of the authorized generics' choice-specific shocks σ_ε, the cost of releasing AG if a trade shock is received $\bar{\kappa}$, and the probability of drawing a trade shock p. We set T = 32 (meaning that the dynamic game lasts 32 quarters) and impose that profits become zero afterwards for reasons such as product obsolescence (the second assumption can be relaxed). The parameters are estimated via simulated method of moments (SMM).

An overview of our estimation procedure is as follows. We follow Coen and Coen (2022) in setting up our objective function, and we adopt their notation and explanation in what follows. For a trial parameter vector $\psi = \{\sigma_\varepsilon, \bar{\kappa}, p\}$, we solve for CCPs of AGs, simulate data, and match the simulated moments with observed data moments. Note that for this part of the estimation procedure, we only need to solve for the CCPs of the AGs, since the equilibrium number of generics is observed in the dataset. See Appendix A.2 for a detailed description of the algorithm.

The estimated parameter vector $\hat{\psi}$ minimizes the sum of squared percentage differences between the theoretical and empirical moments:

$$\hat{\psi} = \arg\min_{\psi} \left(m(\psi) - m_0\right)' \Omega^{-1} \left(m(\psi) - m_0\right)$$

where $\Omega = m_0 m_0'$. We use two sets of trimmed moments for our SMM estimation:

1. Moments about *whether* to release an authorized generic: Our first set of moments captures the variation in decision to release authorized generic versus not across markets. Specifically, the moments are fraction of molecule-formulations that saw AG released, fraction of patient-facing molecule-formulations that saw AG released, fraction of non-patient-facing molecule-formulations that saw AG released, and co-variance of authorized generic release decision with market size

2. Moments about *when* to release the authorized generic: Our second set of moments captures the variation in the timing of authorized generic release. For each molecule-formulation that saw an authorized generic release, we calculate the "authorized generic release delay", i.e. how long after the first generic entrant was the authorized generic released. We use moments of the distribution of authorized generic release delay, specifically its mean, variance, skewness, and kurtosis.

The cost estimation proceeds as follows. The standard errors are estimated via block-bootstrapping, similar to Alam (2023) and Wang (2022).

5.4 Heuristic Discussion on Identification

We now present a brief non-technical discussion on what variation in the data help pin down different parameters. Note that from a technical standpoint, every parameter is identified off of every moment in the data, but certain variations in the data play a bigger role in informing certain parameters.

On the demand side, the price coefficient for Stage 1 (outside option versus brand versus non-brand) is pinned down by how much market share switches from the expensive brand to the cheap non-brand products. That is, the price coefficient captures the correlation between market share gap between brand and nonbrand, and price gap between brand and nonbrand groups. For Stage 2 (within-non-brand competition), the price coefficient is pinned down by how much higher market share is for cheaper non-brand drugs than more expensive non-brand drugs. This setup allows us to have different price sensitivies for the brand-to-nonbrand choice and within-nonbrand choice.

On the supply side, the probability of a trade-shock is pinned down by the variation in authorized generic release decisions across molecule-formulations, i.e. the first set of moments in the SMM estimation. The higher the fraction of molecule-formulations that see authorized generic launch, the higher the probability of the trade-shock. The logit scaling parameter is pinned down by the temporal variation in authorized generic release, i.e. the second set of moments in the SMM estimation. The greater the delay of authorized generic launch after loss-of-exclusivity, the greater the logit scaling parameter.

To capture the aggregate dynamics of a highly complicated US pharmaceutical industry in a stylized and tractable model, we make several simplifications. In this section we discuss some of these modeling choices.

6 Results

Tables 3 and 4 show the results of the demand estimation. From Table 3 we see that price has a negative impact on demand, and that the price-sensitivity is stronger for non-patient-facing molecule-formulations than patient-facing molecule-formulations. We also estimate the random coefficients to have statistically insignificant variance, meaning that any heterogeneity on the demand side is sufficiently picked up by the logit shocks. Table 4 shows that non-brand drugs also face negative price elasticity within the group. Moving to the supply side estimation, we find that brand manufacturers draw a no-trade shock 23.6% of the time and incur negligible costs of entry when they draw a trade-shock.

The fact that the entry cost for trade-shock ($e^{\bar{\kappa}} \approx \$8{,}209$) is almost negligible reflects how the industry works. Most of a generic's entry cost comes from reverse engineering the production process and satisfying the FDA manufacturing requirements. The branded drug manufacturer has already incurred that cost when first developing the product, so there is little cost of releasing the AG. However, there is cost of marketing, manufacturing, and distributing the AG, which is captured by the per-period fixed cost and marginal costs.

It is also possible that the AG faces a smaller per-period fixed cost compared to the

Table 3: Results of demand estimation for Stage 1.

	Patient-facing	Non-patient-facing
ln(price)	-1.004	-2.781
	(0.033)	(0.244)
Brand	0.799	1.033
	(0.069)	(0.250)
Brand * time-since-generic-entry	-0.095	-0.008
	(0.007)	(0.014)
RC std: Brand	-5.946e-07	-1.812e-07
	(6.120)	(6.598)
RC std: Price	3.434e-07	-2.025e-07
	(4.778)	(11.838)

Notes: The demand model is estimated using 2-step nonlinear GMM. Robust standard errors are reported in parentheses. "RC std: Brand" and "RC std: Price" refer to σ_1 and σ_α respectively. Market and quarter fixed effects are suppressed.

Table 4: Results of demand estimation for Stage 2.

	Patient-facing	Non-patient-facing
Authorized Generic	0.857	0.067
	(0.077)	(0.186)
ln(price)	-0.647	-1.286
	(0.023)	(0.068)

Notes: The demand model is estimated using a fixed-effects regression following Maggio et al. (2022). Robust standard errors are reported in parentheses.

generic, since they might use some of the branded drug's existing infrastructure. To capture that possibility, we re-estimate the AG cost distribution with the per-period fixed cost for AGs set at α of generics' per-period fixed cost, where α takes values of 0.6 and 0.8. We continue to get very similar estimates (see Appendix D.1).[27]

[27]Note that it is not the case that the AG faces no per-period fixed cost. Evidence for this comes from the fact that there are many AGs which are outsourced to a generic manufacturer or a specialized AG manufacturer (such as Prasco and Greenstone) for manufacturing, marketing, and distribution. If doing all these were costless, then the brand drug manufacturer would have undertaken the activity by itself. In fact, marketing AGs is a lucrative business that some generic manufacturers try to get.

Table 5: Results of cost estimation.

σ_ε	9.847e+06
	(2.811e+06)
$\bar{\kappa}$	9.013
	(1.691)
p	0.236
	(0.052)

Notes: Estimation done using simulated method of moments. Bootstrapped standard errors are reported in parentheses.

7 Counterfactuals

In this section we first highlight key model dynamics that provide intuition for our subsequent results, then discuss the results of our counterfactuals.

To understand our model and predict the impact of different policies, we solve the model by backward induction. For the counterfactuals, we solve for both the number of generic entrants and CCP of authorized generic. A description of the full-solution method is given in Appendix A.1.

Next we summarize how we report our counterfactual results. We take the model-predicted choice probabilities and run 200 simulations, then report the average results from the simulations. We run the counterfactuals for 222 of the 241 market-specifications (molecule-formulations) found in our data, and study the distribution of effects. Note that we think of counterfactuals under different market specifications as mainly checking how our model predictions vary for different realistic parameter values. We vary generic entry costs and per-period fixed costs across market-specifications; more details of the implementation over market-specifications can be found in Appendix C. Finally, when reporting the simulation results, the term "AG release fraction" denotes the fraction of simulations where an Authorized Generic is released in the market.

We now present a brief discussion on profitability in different phases of a market's lifecycle. The early quarters of a molecule-formulation after loss-of-exclusivity are the most profitable for firms. During these periods, there are few firms in the market, resulting in high markups and high economic profits. As time goes by, more generics get FDA approval and enter the market, resulting in markups being driven close to zero and low economic profits. This provides another reason why generics want to launch earlier – the earlier phase of the market yield the highest profits and so contribute most towards covering the entry cost.

7.1 Deterrence through early-move versus early-launch

In this section, we illustrate how both the independent generics and authorized generic can deter each other's entry, but in different ways. Generics can deter AG release because they make entry decisions earlier; however, the AG can deter generic entry because it is able to launch earlier and provide price competition from the early stages of the market.

An independent generic has an early-move advantage. This is because a generic makes its entry decision and incurs its fixed cost in the first stage, before the authorized generic. This ability to move earlier means that the generic can deter an authorized generic from entering. Note that the authorized generic having a substantially lower entry cost than independent generics (conditional on drawing a trade shock) means that this entry deterrence effect is light.

On the other hand, an authorized generic can deter the independent generic's entry through the fact that it can launch earlier. Since the authorized generic can launch its product early – during the more profitable phase of the market – it has a strong entry deterrence effect on the independent generic.

To showcase these economic forces, we perform the following two counterfactuals.

First, we show that the generics are able to deter AG release when AGs do not enjoy the launch-timing advantage over generics. We fix the following: i) Generics can launch immediately upon loss-of-exclusivity (at $t = 2$), so that the AG no longer enjoys a launch timing advantage over generics; ii) AG has a low logit scaling parameter so that the AG can launch immediately; iii) AGs have the same entry cost as generics, so that the AG no longer has a cost advantage; iv) AG and generics face the same demand primitives, so that neither enjoy any demand-side advantage over the other. Thus, in this counterfactual, the AG no longer has the early-launch advantage or the cost advantage. We find that across nearly all market-specifications, the AG is not launched. The results are summarized in Table 8.

Next, we show that AG is able to deter generics better when it has a stronger early-launch advantage. To do so, we vary the AG's early-launch advantage to see how it affects generic entry. First we run simulations where the AG has low logit scaling parameter, which allows it to launch very early without any delays. Next we run simulations where the AG has a high logit scaling parameter, which means it launches much later due to delays. We find that across most market-specifications, the AG can deter more generics when it can launch earlier. The results are summarized in Table 9.

7.2 Generic approval rates

We examine the effects of faster ANDA (generic) approval rates by the FDA. In our counterfactuals, we scale up the probability of ANDA being approved in every quarter after loss of exclusivity, and simulate the impact on market outcomes.

We find that faster generic approval rates leads to fewer generics entering, prices being lower in initial periods, prices being weakly higher in later periods, and that the effect is more muted if AGs may be present in the market.

Under faster FDA approval rates, there are opposing forces on a generic's entry decision. On the one hand, faster approval rates means the generic can market their drug earlier, and so has a longer time (till the terminal period) to make profits and cover their entry cost. This force would incentivize more generics to enter. On the other hand, faster approval rate means that when a generic does enter the market, more of its competitors are already in the market, so it will face greater competition and smaller profits upon entry.

So which of these forces dominate? This is an open question we can study by simulating the market under faster ANDA approval rates.

For the first group of counterfactuals we assume an authorized generic has drawn a "no-trade shock" and so is not present in the market. For most market specifications (over 80%) we see that as ANDA approval rates increase, there are fewer generic entrants; for the remainder, there is no change in total generics. This means that the entry disincentive from earlier competition generally dominates the entry incentive from longer operation time in the industry.[28] The results are summarized in Table 10. An example for a single

[28] To see why, we can look at the per-period prices (and profits) in the industry. The earliest periods are the most profitable, because there are few nonbrand products in the market, resulting in high markups. As more generics enter, these markups are driven down to zero, so that in the long run price is nearly equal to marginal cost. This means that after a certain number of periods, generic firms no longer make economic profit that can go towards covering their entry cost. With faster ANDA approval rates, price drops to marginal cost much faster, meaning that there are far fewer quarters when generics can make economic profit, because there will

molecule-formulation is given in Table 6.

Next we rerun the counterfactuals but assume that the authorized generic has drawn a "trade shock", meaning it can be released by the incumbent manufacturer. Similar to before, for most market specifications (over 50%), as ANDA approval rates increase, there are fewer generic entrants. However, with AG, the decline in generics as approval rate increases is smaller; that is, as approval rate increases, the number of generics declines by a smaller amount with AG presence compared to without AGs. The results are summarized in Table 11. An example for a single molecule-formulation is given in Table 7.

Table 6: Market outcomes with different generic approval rates for one molecule-formulation. The AG is assumed to have drawn a no-trade shock and therefore cannot enter the market.

Cases	**Total generics**	**Mean generic price (early periods)**	**Mean generic price (later periods)**
Estimated	7.0	3.2	2.1
Estimated x 2	5.0	2.97	2.1
Estimated x 3	4.0	2.85	2.1
Immediate entry	3.0	2.56	2.56

Notes: Simulation conducted for market-specification of molecule-formulation Amoxicilin-Clarithromycin-Lansoprazole-oral. "Estimated x n" refers to the estimated generic approval rates multiplied by n. "Immediate entry" refers to all generics entering the market immediately after loss of exclusivity.

7.3 Ban on Authorized Generic

We find that a ban on AGs leads to weakly more generic entrants – and sometimes multiple additional new generics – but has an ambiguous effect on overall prices. For this section, we assume that the AG would have drawn a "trade shock" had it not been banned, meaning it can be released by the incumbent manufacturer.

In our counterfactuals we find that, when AGs are banned, most market specifications (over 70%) see an increase in the number of generics. In particular, over 40% of the

be more competitors in the market right from the beginning. Thus, while it is true that generics can operate in the market longer due to being approved earlier, most of that time is going to be spent making zero economic profit.

Table 7: Market outcomes with different generic approval rates for one molecule-formulation. The AG is assumed to have drawn a trade shock and therefore can enter the market.

Cases	Total generics	AG release fraction	Mean generic price (early periods)	Mean generic price (later periods)	Mean AG price (early periods)	Mean AG price (later periods)
Estimated	6.0	1.0	3.16	2.1	2.74	2.1
Estimated x 2	4.0	1.0	3.04	2.1	2.84	2.1
Estimated x 3	4.0	1.0	2.76	2.1	2.56	2.1
Immediate entry	3.0	1.0	2.48	2.1	2.1	2.1

Notes: Simulation conducted for market-specification of molecule-formulation Amoxicilin-Clarithromycin-Lansoprazole-oral. "Estimated x *n*" refers to the estimated generic approval rates multiplied by *n*. "Immediate entry" refers to all generics entering the market immediately after loss of exclusivity.

markets see multiple additional new generics entering. However, less than 30% market specifications do not see any new generics when AG is banned. For a majority of market-specifications, AG ban leads to higher prices in the early periods of the market. If the AG was deterring multiple generics from entering, prices are lower in the later periods of the market with an AG ban. The exact effect on number of generics and prices depends on the specific demand and supply parameters. The results are summarized in Table 12.[29]

We first explain the impact from an AG ban on generic entrants. There are two opposing forces operating on different phases of the market's timeline after loss-of-exclusivity:

1. Early AG launch: Without an AG ban, the AG is released early in the market, so there is always an additional competitor in the market from the early quarters. Without the AG, the generics face lower competition in the initial periods as all the generics

[29] We also simulate the impact of banning AG entry only for the first few periods (1-2 quarters). We find that the ban leads to smaller generic deterrence. The results are summarized in Table 13.

slowly gain approval. Therefore, under an AG ban, prices are generally higher in the early phase of the market.

2. Greater generic entry: An AG ban leads to greater number of generic entrants. More generic manufacturers in the market drive prices down in the later phase of market.

An AG ban generally leads to higher prices in the early periods of the market. However, for a minority of market-specifications, we find that the AG ban could lead to lower prices. These are markets where the AG chooses to launch later than other markets, so that the AG ban has minimal effect on early-period competition (as the AG is not available during those times anyways).

The effect on later-period prices depends (among other things) on how many generics are deterred by the presence of an authorized generic. If an authorized generic ban leads to multiple new generics entering, then the drop in price in the later phase of the market more than offsets the missing price competition from the AG (Table 15). As a result, in these markets the AG ban leads to lower prices in later periods. However if an AG ban leads to one or no new additional generic entrants, then AG ban has minimial effect on prices in the later phase of the market (Table 14).

Why can an AG ban lead to more than one new generic entering? This is because the AG is a more potent competitor than independent generics as a result of being released earlier in the market. Since the AG is released early, it has a negative impact on every generic and particularly during the most profitable early periods of the market. If the AG is banned, not only is there one fewer competitor, but it is also one less competitor in the most profitable periods in the market. This results in a significant increase in expected generic profits compared to one fewer independent generic. In turn, this can incentivize multiple new generics to enter than if an AG had been present.

Why can an AG ban lead to no additional generics entering? While the AG ban makes the market more profitable, an additional generic may still not anticipate enough expected profit to cover its cost. Note that, conditional on receiving a trade shock, an AG has lower entry cost than an independent generic. Furthermore, the AG can launch immediately, and so raise profits over a longer period to cover its entry costs. This means that just because an AG found it profitable to enter a market does not mean that an independent generic would also have entered that market.

How many independent generics are deterred by the presence of an authorized generic also depends on market-specific factors, particularly generic entry costs.

7.4 Discussion on No-AG settlements

No-AG settlement are struck by the brand manufacturer to delay the first generic entrant, resulting in the exclusivity period lasting longer and consumers paying high prices over this extended time. See Section 4.3 and FTC (2011) for an in-depth discussion.[30] Due to lack of data, we do not seek to model no-AG settlements and their welfare implications. Instead, we offer a brief discussion on the implications of no-AG settlements through the lens of our model.

Generally, no-AG settlement means less competition at the beginning of loss-of-exclusivity – the generics entering in the first few quarters do not face an additional competitor in the authorized generic, resulting in higher prices.

However, the negative effects of a no-AG settlement can be partially negated by higher generic entry – no AG means one less competitor in the market, leading to more generics deciding to enter the molecule-formulation. The exact effect of a no-AG settlement on generic entry can therefore depend on how early the generic manufacturers find out that such a deal is in place. Let us suppose that the market conditions were favorable for an authorized generic to be released in the absence of a no-AG settlement. If generics were expecting an authorized generic to enter, then some of them would not apply for ANDA to that market. However if then a no-AG deal is struck, there will be fewer generics, at least in the short term. Potential entrants realize an authorized generic will not in fact be released and file for ANDA approval, resulting in delayed generic entry than otherwise. If generics realize no-AG settlement has been struck early enough, then they can apply for ANDA earlier, resulting in earlier product launch. One policy proposal could therefore be that the brand manufacturer needs to immediately announce when a no-AG settlement has been signed, which will allow additional generics to start applying for entry.

The above discussion highlights the interesting dynamics behind no-AG settlement and their welfare discussions. Furthermore, no-AG settlement may be more complex than de-

[30]The FTC also filed an amicus brief regarding no-AG settlements, see https://www.ftc.gov/legal-library/browse/amicus-briefs/re-effexor-xr-antitrust-litigation-0

scribed above; see Bokhari et al. (2020) for a discussion. In short, this is potentially an exciting avenue for future research with antitrust implications.

8 Conclusion

In the pharmaceutical industry, branded drug manufacturers can compete with generics by releasing an Authorized Generic, which is identical to the branded drug but without the brand label attached. Using quarterly drug sales and revenue data on US for 2004-2016, we estimate a structural model of pharmaceutical demand, generic entry, AG release, and pricing. We first set up and estimate a discrete choice model of demand where the consumer first chooses a group (brand, nonbrand, or outside option) then chooses a product within the group. Next we build a two-stage supply model of static generic entry followed by a dynamic AG release timing game. The model is solved using backward induction to recover equilibrium generic entrants and conditional choice probabilities of AG release. We estimate this model using simulated method of moments to recover approximate bounds on the per-period fixed cost and the AG entry cost distribution.

We simulate market equilibrium under different environments. First, we show how deterrence works in this setting - generics deter through being able to file for entry earlier (early-mover advantage) while AGs can deter because they can launch the product earlier. Next, we find that a faster approval rate leads to weakly fewer generic entrants. This is because greater competition upon launch offsets the advantage from an early launch. Finally, we show that a ban on authorized generics leads to weakly more generic entrants and ambiguous effect on overall prices. The latter happens because while there is increased price competition from weakly more generic entrants in the later stages of the market, it is counteracted by lack of an additional competitor through the AG in the early stages of the market. The exact effect depends on how many generics is being deterred by the AG; for a large number of market specifications, we find that the AG may deter multiple generics.

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A Model solution and cost estimation

A.1 Solving the full model

We solve the model by backward induction and recover the equilibrium number of generic entrants and CCPs of authorized generics.

Note that the AG manufacturer knows the total generics approved in t = 2, i.e. $\mathscr{N}_2$, but can only form probabilistic beliefs about future generic launches. In contrast, generic manufacturers only know the equilibrium generic entrants n_e^*, not $\mathscr{N}_2$.

However, the CCP of the AG varies based on the exact value of $\mathscr{N}_2$, since they suggest different probability distributions over future generic launches. That is, the CCP of AG varies by $\mathscr{N}_2$.

The CCP of the AG is used to calculate the value function of the generics in stage 1. While the generics don't know the exact value of $\mathscr{N}_2$, they can form probabilistic beliefs about it using the FDA approval rates.

Thus, for the model solution, first we solve for the CCP of the AG and construct the corresponding generic value function for each $\mathscr{N}_2$. Then each generic value function (at the initial state s_1) is weighted by the probability of the corresponding $\mathscr{N}_2$ being realized. The resulting weighted generic value function is then used for the first stage.

The solution method is as follows:

1. For a given number of equilibrium generic entrants $\bar{n}$:
   (a) For each $\mathscr{N}_2$ between 0 to $\bar{n}$:
      i. Calculate the authorized generic's state transition matrices as determined by $\bar{n}$, T, FDA approval rates, and $\mathscr{N}_2$.
      ii. Solve for conditional choice probabilities (CCPs) of authorized generic by backward induction from terminal period T. The CCP gives the probability of AG being released in every state of the world from $t = 1$ to T.
      iii. Use the CCP, (12), T, and FDA approval rates to calculate the state transition matrices for generics.

iv. These state transition matrices are combined with per-period profits to construct value functions of generics.

v. Separately, calculate the probability of this $\mathcal{N}_2$ being realized using data on FDA approval rates.

(b) For the initial state s_1, find the generic value function by taking the probability-weighted average of the generic value functions for each $\mathcal{N}_2$ at the state s_1. The probability of each $\mathcal{N}_2$ is being used as weights in this weighted average.

(c) Denote this value function as $V(\bar{n})$ (other arguments suppressed).

2. Repeat the above steps for $\bar{n} = 1, \ldots N$ where N is a large integer.

3. The equilibrium number of generics n^* is such that $V(n^*) \geq \kappa_m^g$ and $V(n^* + 1) < \kappa_m^g$. The value function is decreasing in $\bar{n}$.

4. For this n^* re-solve the model to get the equilibrium CCPs of authorized generics conditional on n^* and every possible $\mathcal{N}_2$.

Note that during simulations for a given market, we run many simulations for each possible $\mathcal{N}_2$, then report probability-weighted averages of each summary statistic (such as AG release fraction, AG release time, mean prices, etc.). The probability of each $\mathcal{N}_2$ is being used as weights in this weighted average. We rescale probabilities if some $\mathcal{N}_2$ see no AG released.

A.2 AG entry cost estimation

For AG cost estimation, we can skip a large chunk of the full solution method. From the data we observe n_{em}^* for each market m as well as $\mathcal{N}_{m2}$.

For a given value of parameters:

1. We solve for conditional choice probabilities (CCPs) of authorized generic by backward induction from terminal period T. The CCP gives the probability of AG being released in every state of the world from $t = 1$ to T.

2. The estimated CCPs are used to run S simulations for each molecule-formulation. In each simulation, generics are randomly released in the market based on Equation

(12) and estimated generic approval rate. The authorized generic is released based on the estimated CCP.

3. We calculate moments of the simulated data. These include summary statistics of how often the authorized generic is released and delay after loss of exclusivity.

4. The simulated moments are used to calculate the objective function. Intuitively, the objective function gives a scalar measurement of how far the simulated moments are from the data moments.

This process is repeated for different trial values of parameters until the objective function is minimized.

Note that the data moments are covariances over observed releases and market characteristics. Our simulations give us the AG release fraction for each market (i.e. for many simulations at a single market, what fraction of simulations saw AG released). To get the data moment and simulation moment to align, we calculate the simulated AG release fraction for each market, then create a simulated dataset where we duplicate each market many times and assign AG release indicators to them, with the proportion of AG release indicators correspond to the calculated AG release fraction for that market. We do this for each market, then take the covariance of AG release indicators and market characteristics in this simulated dataset. This gives us the simulated moment.

The standard errors are estimated via block-bootstrapping, similar to Alam (2023) and Wang (2022). In what follows we closely adopt the explanation from Alam (2023). We sample molecule-formulations from the data with replacement until we match the total number of molecule-formulations in our full dataset. Then we re-estimate the model parameters for each bootstrap sample. This is used to construct the sampling distribution of each parameter.

A.3 Estimation of per-period fixed costs

Fix some trial value of generic entry cost κ_g across all molecule-formulations. We try $\kappa_g =$ \$3 million, \$10 million, and \$15 million. (Intuitively, the \$3 and \$15 million provide upper and lower bounds on the per-period fixed cost respectively, as we explain below.) For a given value of κ_g, we pick a grid of values for per-period fixed cost. For each per-period fixed cost on the grid, solve for the equilibrium number of generic entrants.

The objective function is the sum of squared differences between the observed generic entrants in each market and the predicted entrants. One issue is that the model cannot predict less than zero entrants for a market (i.e. the outcome variable is censored at 0) and so the objective function may be minimized by predicting zero entrants for lots of markets. To tackle this problem, the objective function also includes a "censoring penalty", where if a market is predicted to have zero generic entrants, it is assigned the censoring penalty value instead of the difference between observed and predicted entrants. This is to prevent the estimation from assigning zero entrants to lots of markets to minimize the objective function. In our estimation, we try censoring penalties of 3 and 4 and obtain similar results.

Returning to the estimation, for a given κ_g, we calculate the objective function for every value of per-period fixed cost on the grid. We pick the value that minimizes the objective function. This gives us the per-period fixed cost for that κ_g. We repeat this process for the different κ_g mentioned previously to get the per-period fixed cost for each κ_g.

The grids are spaced from \$0 to \$4 million at intervals of \$100,000. The grid is chosen to be wide enough so as to not get a corner solution, and fine enough to try to ensure that a global minimum not missed. The objective function is non-convex.

B Derivation of Product-Level Market Share

We now derive the expression for product-level market share. We follow the notation of Conlon and Gortmaker (2020). Our derivation follows similar derivations in Berry et al. (1995), Conlon and Gortmaker (2020), Nevo (2000), and other related papers. Throughout the derivation, the time subscript is suppressed for clarity.

We can rewrite Stage 1 in terms of mean utilities and match values following Berry et al. (1995).

$$
\begin{aligned}
u_{ig} = {} & \gamma_m^{(1)} + \lambda_t^{(1)} + \alpha_i^{(1)} ln\, p_g + \beta_{1i}^{(1)}\text{brand}_g + \beta_2^{(1)} brand_g \cdot \text{time-since-loe} + \xi_g^{(1)} + \varepsilon_{ig} \\
& = \gamma_m^{(1)} + \lambda_t^{(1)} + \alpha^{(1)} ln\, p_{gt} + \beta_{1i}^{(1)}\text{brand}_g + \beta_2^{(1)} brand_g \cdot \text{time-since-loe} + \xi_g^{(1)} + \\
& \qquad \sigma_\alpha \nu_{\alpha,i} ln\, p_g + \sigma_\beta \nu_{\beta,i}\text{brand}_g + \varepsilon_{ig} \\
& = \delta_g^{(1)} + \mu_{ig} + \varepsilon_{ig}
\end{aligned}
$$

where the mean utility of group g, $\delta_g^{(1)}$, is given by:

$$\delta_g^{(1)} = \gamma_m^{(1)} + \lambda_t^{(1)} + \alpha^{(1)} ln\, p_g + \beta_1^{(1)}\text{brand}_g + \beta_2^{(1)} brand_g \cdot \text{time-since-loe} + \xi_g^{(1)}$$

and the match value of individual i with group g, μ_{ig}, is given by:

$$\mu_{ig} = \sigma_\alpha \nu_{\alpha,i} ln\, p_g + \sigma_\beta \nu_{\beta,i}\text{brand}_g$$

Let the idiosyncratic shocks ε_{ig} and ε_{ij} follow a Type-1 Extreme Value distribution. Then the probability of consumer i choosing group g is given as:

$$s_{ig} = \frac{e^{\delta_g^{(1)} + \mu_{ig}}}{\sum_l e^{\delta_l^{(1)} + \mu_{il}}}$$

For individual i, let μ_i be the vector of her match-values for all groups in the market, i.e. if both brand and nonbrand drugs are in market, μ_i is 3×1. Similarly, let δ_i be the vector of her mean-values for all products in the market. Since there are many consumers with heterogeneous preferences (reflected in the random coefficients $\alpha_i^{(1)}$ and $\beta_i^{(1)}$), the market share of group g is given by integrating over the distribution of random coefficients:

$$s_g = \int \frac{e^{\delta_g^{(1)} + \mu_{ig}}}{\sum_l e^{\delta_l^{(1)} + \mu_{il}}} f(\mu_i | \tilde{\theta}_2) d\mu_i$$

where $\theta_2 = (\sigma_\alpha, \sigma_\beta)$

Conditional on choosing nonbrand group, the probability of a consumer choosing product j from nonbrand group is given by:

$$s_{j|g} = \frac{e^{\delta_j^{(2)}}}{\sum_k e^{\delta_k^{(2)}}}$$

where $\delta_k^{(2)} = \gamma_{m(k)}^{(2)} + \beta^{(2)}\text{AG}_k + \alpha^{(2)} ln(p_k) + \xi_k^{(2)}$.

Stage 1 gives the expression for the market share of a group, while Stage 2 gives the expression for the market share of a product conditional on a group. Therefore, we can express the market share of a single good in terms of group market share and market share of good conditional on choice of group.

$$d_{ij} = \begin{cases} 1 & \text{if } U_{ij} > U_{ik} \text{ for all } k \neq j, \\ 0 & \text{otherwise.} \end{cases}$$

Therefore we can express the market share of a single good in terms of group market share and market share of good conditional on choice of group:

$$\begin{aligned} s_j =& \int d_{ij}(\delta + \mu_i) d\mu_i d\varepsilon_i \\ =& \int \frac{e^{\delta_g + \mu_{ig}}}{\sum_l e^{\delta_l + \mu_{il}}} \frac{e^{\delta_j + \mu_{ij}}}{\sum_k e^{\delta_k + \mu_{ik}}} f(\mu_i|\tilde{\theta}_2) d\mu_i \\ =& \int \frac{e^{\delta_g + \mu_{ig}}}{\sum_l e^{\delta_l + \mu_{il}}} \frac{e^{\delta_j}}{\sum_k e^{\delta_k}} f(\mu_i|\tilde{\theta}_2) d\mu_i \\ =& \left[\int \frac{e^{\delta_g + \mu_{ig}}}{\sum_k e^{\delta_l + \mu_{il}}} f(\mu_i|\tilde{\theta}_2) d\mu_i \right] \frac{e^{\delta_j}}{\sum_k e^{\delta_k}} \\ =& s_{g(j)} s_{j|g(j)} \end{aligned}$$

Therefore,

$$s_j = s_{g(j)} s_{j|g(j)}$$

C Counterfactual results

For implementing the counterfactuals, we divide the 241 molecule-formulations into "very small", "small", "medium", and "large" markets. The "very small" markets are assigned per-period fixed cost of and generic entry cost of , "very small" markets are assigned per-period fixed cost of and generic entry cost of , "medium" markets are assigned per-period fixed cost of and generic entry cost of , and "large" markets are assigned per-period fixed cost of and generic entry cost of .

Now we explain how we classify markets into these 4 groups. The general problem is that there is variability in generic entry costs and potentially per-period fixed costs across molecule-formulations; however we only have approximate bounds for each. Reported generic entry costs vary from \$3 to \$15 million. Our approximate bounds on fixed costs vary from \$200,000 to \$ 2.5 million. When we calculate predicted generic entrants at midpoints of these bounds (\$10 million generic entry cost and \$1.2 million per-period fixed costs) then a large number of molecule formulations predict 0 generic entrants or more than 22 generic entrants (the highest number observed in our dataset).

As a result, for each molecule-formulation, we try the 4 sets of generic entry costs and per-peroid fixed costs mentioned previously. A molecule-formulation is assigned to the group with the largest costs such that the model predicts positive generic entrants less than 22.

For 19 of the 241 molecule-formulations our model does not predict positive generic entrants fewer than 22, and we drop those markets from our counterfactuals.

Note that we think of counterfactuals under different market specifications as mainly checking how our model predictions vary for different realistic parameter values.

A note on terminology: throughout this section "Early periods" refers to first 4 quarters, and "Later periods" refers to last 4 quarters (the terminal period is quarter 32).

C.1 Early-move vs early-launch

Table 8: Distribution of outcomes across market-specifications.

Percentiles	AG release fraction
10%	0.0
20%	0.0
30%	0.0
40%	0.0
50%	0.0
60%	0.0
70%	0.0
80%	0.0
90%	1.0

Notes: For this counterfactual we fix the following: i) Generics can launch immediately upon loss-of-exclusivity (at $t = 2$), so that the AG no longer enjoys a launch timing advantage over generics; ii) AG has a low logit scaling parameter so that the AG can launch immediately; iii) AGs have the same entry cost as generics, so that the AG no longer has a cost advantage; iv) AG and generics face the same demand primitives, so that neither enjoy any demand-side advantage over the other. We find that AG almost never launches.

Table 9: Distribution of outcomes across market-specifications.

Percentiles	**Difference in generics**
10%	-2.0
20%	-2.0
30%	-2.0
40%	-2.0
50%	-2.0
60%	-1.0
70%	-1.0
80%	-1.0
90%	0.0

Notes: We vary the AG's early-launch advantage to see how it affects generic entry. First we run simulations where the AG has low logit scaling parameter, which allows it to launch very early without any delays. Next we run simulations where the AG has a high logit scaling parameter, which means it launches much later due to delays. "Difference in generics" is the equilibrium generic entrants when the AG launches early minus the equilibrium generic entrants when the AG launches late. We limit to market-specifications where AG launches at least once across all simulations for both low and high logit scaling parameters. We find that across most market-specifications, the AG can deter more generics when it can launch earlier.

C.2 Faster generic approval rates

Table 10: Distribution of outcomes across market-specifications with estimated versus immediate generic approval. The AG is assumed to have drawn a no-trade shock and therefore cannot enter.

Percentiles	**Difference in no. of generics**	**Difference in generics price (early periods)**	**Difference in generics price (later periods)**
10%	0.0	0.174	-5.267
20%	0.0	0.332	-1.373
30%	1.0	0.503	-0.573
40%	1.0	0.734	-0.306
50%	1.0	1.22	-0.157
60%	2.0	1.958	-0.069
70%	2.0	5.276	-0.031
80%	3.0	9.751	-0.0
90%	4.0	36.129	0.0

Notes: "Difference in no. of generics" is the equilibrium generic entrants under estimated generic approval rates minus the equilibrium generic entrants under immediate generic approval. Other variables similarly defined. We find that faster generic approval rates lead to fewer generics entering the market.

Table 11: Distribution of outcomes across market-specifications with estimated versus immediate generic approval. The AG is assumed to have drawn a trade shock and therefore can enter.

Percentiles	**Difference in no. of generics**	**Difference in generics price (early periods)**	**Difference in generics price (later periods)**
10%	0.0	0.003	-2.533
20%	0.0	0.189	-0.446
30%	0.0	0.344	-0.105
40%	0.0	0.492	-0.038
50%	1.0	0.824	-0.0
60%	1.0	1.499	-0.0
70%	1.0	2.739	0.0
80%	2.0	8.077	0.0
90%	3.0	21.498	0.004

Notes: "Difference in no. of generics" is the equilibrium generic entrants under estimated generic approval rates minus the equilibrium generic entrants under immediate generic approval. Other variables similarly defined. We find that faster generic approval rates lead to fewer generics entering the market.

C.3 Ban on AG

Table 12: Distribution of outcomes across market-specifications.

Percentiles	**Difference in no. of generics**	**Difference in generics price (early periods)**	**Difference in generics price (later periods)**
10%	-3.0	-1.697	-0.594
20%	-3.0	-0.441	-0.001
30%	-2.0	-0.166	0.0
40%	-2.0	-0.071	0.0
50%	-1.0	-0.031	0.011
60%	-1.0	-0.011	0.05
70%	-1.0	0.0	0.143
80%	0.0	0.037	0.369
90%	0.0	0.184	1.298

Notes: "Difference in no. of generics" is the equilibrium generic entrants when AG is allowed minus equilibrium generic entrants when AG is banned. Other variables similarly defined. We find that banning AG generally leads to more generic entrants. We assume that in the absence of a ban, the AG would have drawn a trade shock and therefore could have chosen to enter.

Table 13: Distribution of outcomes across market-specifications.

Percentiles	**Difference in no. of generics**	**Difference in mean AG release time (qtrs)**
10%	-2.0	-6.626
20%	-1.0	-5.548
30%	-1.0	-4.964
40%	-1.0	-4.566
50%	-1.0	-4.185
60%	0.0	-3.992
70%	0.0	-3.884
80%	0.0	-3.568
90%	0.0	-2.321

Notes: "Difference in no. of generics" is the equilibrium generic entrants when AG is allowed minus equilibrium generic entrants when AG is banned for the first 2 quarters. Other variables similarly defined.

Table 14: Distribution of outcomes across market-specifications.

Percentiles	**Difference in generics price (early periods)**	**Difference in generics price (later periods)**
10%	-1.731	-3.531
20%	-0.603	-0.196
30%	-0.276	-0.001
40%	-0.109	0.0
50%	-0.034	0.0
60%	-0.014	0.0
70%	0.0	0.0
80%	0.02	0.011
90%	0.149	0.09

Notes: From Table 13 we isolate the market-specifications that see less than two additional generics when AG is banned. This table shows that in these market-specifications, banning AG generally leads to higher generic prices in the early periods and little effect on prices in later periods. "Early periods" refers to first 4 quarters, and "Later periods" refers to last 4 quarters (the terminal period is quarter 32).

Table 15: Distribution of outcomes across market-specifications.

Percentiles	**Difference in generics price (early periods)**	**Difference in generics price (later periods)**
10%	-0.966	0.011
20%	-0.276	0.041
30%	-0.092	0.072
40%	-0.054	0.117
50%	-0.026	0.242
60%	-0.009	0.34
70%	0.011	0.449
80%	0.06	1.072
90%	0.204	2.269

Notes: From Table 13 we isolate the market-specifications that see more than two additional generics when AG is banned. This table shows that in these market-specifications, banning AG generally leads to higher generic prices in the early periods and but lower prices in later periods. "Early periods" refers to first 4 quarters, and "Later periods" refers to last 4 quarters (the terminal period is quarter 32).

D Additional estimation results

D.1 Cost estimation

E Additional summary statistics

Delay	Count
0.0	58
1.0	10
2.0	6
3.0	2
4.0	2
5.0	3
6.0	2
8.0	1
9.0	1
10.0	1
11.0	1
12.0	2
14.0	2
15.0	2
17.0	1
19.0	2
20.0	2
21.0	1
22.0	2
27.0	1
38.0	1
41.0	1

Table 16: AG Delay table. This shows, of the 110 AGs released, how many quarters after loss-of-exclusivity they were released. "Delay" is the number of quarters after loss-of-exclusivity. "Count" is the number of AGs released for the corresponding quarters since loss-of-exclusivity.

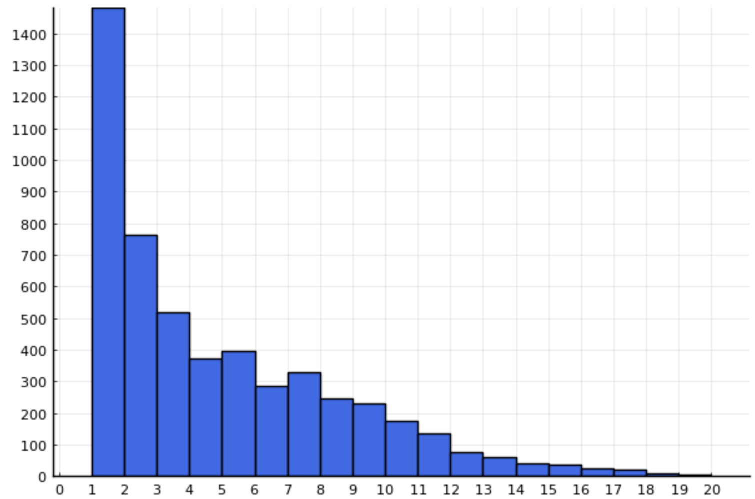

Figure 5: Histogram of count of generics present by molecule-formulation-quarter.

F Miscellaneous

F.1 Comparison with Fowler et al. (2023)

One difference between Fowler et al. (2023) and our findings is they find AG incidence is just 6% of the markets, while we find nearly 44%. We believe this is mainly because they define product and possible markets for AG release ("at risk markets") more broadly than us. In particular, differences in AG entry rate arise due to:

1. Definition of product: We consider a product market to be "molecule-formulation", where formulation is either oral, injectable, or everything else. Fowler et al. (2023) define a product to vary by strength and more detailed dosage forms. As a result, their number of markets is much larger than ours.

2. At-risk markets: We consider a market to potentially see AG entry (at-risk market) if we find that it has seen at least one generic entry (and that first generic entry happened after 2004). Instead, Fowler et al. (2023) considers at-risk products to be "number of products approved during or after 1985 that had not been withdrawn or discontinued and that still had patent or exclusivity protection in each year from 2010 to 2019". This is a very broad definition. Since AG release is only considered when a generic enters, our definition is consistent with our reesarch question.

3. Undercounting AGs: Fowler et al. (2023) mentions that for data issues "... led to conservative estimates of the incidence of authorized generic entry"
4. Difference in yearly coverage: We look at 2004-2016, while they consider 2010-2019.

Other findings from Fowler et al. (2023) match our paper closely.